



RAMAIAH
Institute of Technology

Project Report on

**Design and Implementation of an IoT-Enabled Smart
Sugarcane Farming System**

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CERTIFICATE

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DECLARATION

We hereby declare that the project entitled “Design and Implementation of an IoT-Enabled Smart Sugarcane Farming System” has been carried out independently by us, under the guidance of Dr. S G Shivaprasad Yadav, Associate Professor, Electronics & Telecommunication Engineering, Ramaiah Institute of Technology, Bangalore. This report has been submitted in partial fulfilment for the award of degree, Bachelor of Engineering in Electronics & Telecommunication Engineering of Ramaiah Institute of Technology (Autonomous Institute, affiliated to VTU, Belgaum) during the year 2025-2026.

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ABSTRACT

Agriculture plays a crucial role in the economy, and efficient monitoring of crop conditions is essential for improving productivity and reducing resource wastage. This project presents an IoT Enabled Smart Sugarcane Farming System designed to monitor environmental and soil parameters in real time using wireless sensor networks and cloud technologies. The system consists of three sensor nodes and one central gateway. Each sensor node uses an STM32 Nucleo F401RE microcontroller integrated with a DHT11 sensor, capacitive soil moisture sensor, rain sensor, and SX1278 LoRa module for long-range wireless communication.

The collected sensor data is transmitted through LoRa communication to an ESP32-based gateway, which displays the received data on an I2C LCD and forwards it to a remote monitoring system through Wi-Fi. The gateway also interfaces with solenoid valves and a water pump through a relay module for irrigation purposes. Unlike fully automated systems, irrigation control is manually operated through the Blynk mobile application, allowing farmers to make irrigation decisions based on real-time moisture levels and the growth requirements of sugarcane at different stages.

For remote monitoring, a web-based dashboard integrated with a Supabase cloud database is used to display live sensor readings, historical data trends, and smart alerts with precautionary recommendations. The proposed system helps farmers monitor field conditions efficiently, optimize water usage, and support informed decision-making, thereby contributing to improved agricultural productivity and sustainable farming practices.

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LIST OF ACRONYMS

1. IoT	Internet of Things
2. STM32	STMicroelectronics 32-bit Microcontroller
3. ESP32	Espressif 32-bit Microcontroller with Wi-Fi and Bluetooth
4. LoRa	Long Range Communication
5. LCD	Liquid Crystal Display
6. I2C	Inter-Integrated Circuit
7. RTC	Real-Time Clock
8. GPIO	General Purpose Input Output
9. SPI	Serial Peripheral Interface
10. UART	Universal Asynchronous Receiver Transmitter
11. ADC	Analog to Digital Converter
12. DHT	Digital Humidity and Temperature Sensor
13. MCU	Microcontroller Unit
14. REST API	Representational State Transfer Application Programming Interface
15. JSON	JavaScript Object Notation
16. HTTP	HyperText Transfer Protocol
17. DB	Database
18. UI	User Interface

Project Outcome Mapping

1. Review the literature and identify a suitable problem by analyzing the requirements based on current trends and societal needs in the domain of interest and arrive at the specifications. (POs 1, 2, 3,4, 6, 7, 9,11,12) (PSO 1, 2, 3)
2. Identify the clear objectives and methodology for implementing the project by visualizing the Hardware and Software. (POs 1, 2, 3, 4,6, 7, 9, 11, 12) (PSO 1, 2, 3)
3. Design and Implementation of identified Problem using appropriate modern tools and Techniques in the area of Telecommunication/ multidisciplinary areas.(POs 1, 2, 3, 4, 5, 6, 7, 9, 11, 12) (PSO 1, 2, 3)
4. Validate the achieved results and demonstrate good project defense, presentation skills, leadership and punctuality as a team/individual.(POs 1, 2, 3,4, 8, 9, 10, 11, 12) (PSO 1,3)
5. Ability to write the thesis following ethical values and publish the work in quality conferences/journals supporting lifelong learning abilities. (POs 8, 9, 10, 12) (PSO 3)

Mapping of Course outcome to Program outcome

Course Outcomes	Program Outcomes														
	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2	PSO3
CO1	3	3	2	3		2	2		3		2	2	3	2	1
CO2	3	3	3	3		2	2		3		3	2	3	2	2
CO3	3	3	3	3	3	2	2		3		3	2	3	2	2
CO4	3	3	3	3				3	3	3	3	1	2	-	3
CO5								3	3	3		3	-	-	3
Overall	3	3	2.75	3	3	2	2	3	3	3	2.75	2	2.75	2	2.2

Program Outcomes

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Design and Implementation of an IoT-Enabled Smart Sugarcane Farming System

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Justification:

1. CO1 mapped to PO1, PO2, PO3, PO4, PO6, PO7, PO9, PO11, PO12 and PSO1, PSO2, PSO3

CO1: Review the literature and identify a suitable problem by analyzing the requirements based on current trends and societal needs in the domain of interest and arrive at the specifications.

Justification:

The project involved extensive literature review on IoT-based smart agriculture systems, LoRa communication, cloud monitoring platforms, and irrigation management systems. Engineering fundamentals and domain knowledge were applied to analyze real agricultural problems such as inefficient irrigation, water wastage, and lack of real-time monitoring in sugarcane farming. The team studied existing systems, identified limitations, and finalized system specifications based on practical agricultural requirements. The project also considered environmental sustainability through efficient water management and involved teamwork, project planning, and continuous learning of modern agricultural technologies and IoT systems.

2. CO2 mapped to PO1, PO2, PO3, PO4, PO6, PO7, PO9, PO11, PO12 and PSO1, PSO2, PSO3

CO2: Identify the clear objectives and methodology for implementing the project by visualizing the Hardware and Software.

Justification:

The project required designing the complete methodology and architecture of the IoT-enabled smart farming system. Hardware components such as STM32 microcontrollers, ESP32 gateway, LoRa modules, sensors, relay modules, solenoid valves, and power management circuits were selected and integrated. Software tools including STM32CubeIDE, Arduino IDE, Supabase cloud database, web dashboard, and Blynk application were implemented. The methodology involved communication system design, cloud integration, data acquisition, irrigation control, and power optimization. The work addressed societal and environmental issues by improving irrigation efficiency and supporting sustainable agriculture through teamwork and proper project management.

3. CO3 mapped to PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO11, PO12 and PSO1, PSO2, PSO3

CO3: Design and Implementation of identified Problem using appropriate modern tools and Techniques in the area of Telecommunication/multidisciplinary areas.

Justification:

The project involved the practical implementation of an IoT-based smart agriculture system using modern engineering tools and telecommunication technologies. LoRa wireless communication was implemented for long-range data transmission between sensor nodes and the ESP32 gateway. Modern embedded development tools, cloud technologies, REST APIs, web dashboards, and mobile applications were used for real-time monitoring and irrigation support. Experimental investigations, testing, and validation were carried out under real agricultural field conditions. The system promoted sustainable water utilization and multidisciplinary integration of embedded systems, IoT, wireless communication, cloud computing, and agriculture technologies.

4. CO4 mapped to PO1, PO2, PO3, PO4, PO8, PO9, PO10, PO11, PO12 and PSO1, PSO3

CO4: Validate the achieved results and demonstrate good project defense, presentation skills, leadership and punctuality as a team/individual.

Justification:

The implemented system was tested and validated through real-time deployment in sugarcane fields. Sensor readings, LoRa communication reliability, cloud monitoring, irrigation control, and power management performance were analyzed and verified. The project required effective teamwork, leadership, communication, and technical presentation skills during reviews and demonstrations. Ethical engineering practices were followed while designing a system beneficial to farmers and society. Proper project planning, coordination, documentation, and timely completion of tasks helped improve management and professional skills along with lifelong learning capabilities.

5. CO5 mapped to PO8, PO9, PO10, PO12 and PSO3

CO5: Ability to write the thesis following ethical values and publish the work in quality conferences/journals supporting lifelong learning abilities.

Justification:

The project report was prepared following proper academic and ethical standards including literature citation, technical documentation, and systematic presentation of results. The team developed professional report writing, technical communication, and presentation skills while documenting hardware design, software implementation, testing procedures, and experimental results. Team coordination and collaborative work improved interpersonal skills. The project also encouraged self-learning and continuous improvement in emerging technologies such as IoT, wireless communication, cloud computing, and smart agriculture systems, thereby supporting lifelong learning abilities.

CHAPTER - 1

INTRODUCTION

This chapter introduces the challenges in conventional sugarcane farming and the importance of real-time monitoring in improving crop productivity and efficient water management. It explains the motivation behind the project and the need for an IoT-based smart farming solution. The chapter also provides an overview of the proposed system, its applications, and the limitations of the existing approach.

1.1 Need for IoT Enabled Smart Sugarcane Farming System

Agriculture is one of the major sectors that contributes significantly to the economy and plays an important role in meeting food and industrial demands. Sugarcane is one of the most widely cultivated crops and serves as a primary raw material for sugar production and several other by-products. Since sugarcane is a water-intensive crop, maintaining proper environmental and soil conditions is essential throughout its growth period. Figure 1.1 shows the sugarcane field at the VC Farms, Mandya where the proposed IOT systems was tested. Parameters such as soil moisture, temperature, humidity, and rainfall directly influence crop growth, productivity, and quality. Any variation in these conditions can adversely affect crop development and lead to reduced yield.

Traditional farming methods generally rely on manual inspection and experience-based decisions for irrigation and field management. Farmers need to physically visit agricultural fields to observe crop conditions and determine irrigation requirements. This process consumes considerable time and labour and may not always provide accurate information regarding actual field conditions. Excessive irrigation results in water wastage and may cause waterlogging, while insufficient irrigation can lead to moisture stress and reduced crop growth. Such practices affect productivity and increase operational costs.

With advancements in Internet of Things (IoT) technology, sensor-based monitoring systems can be employed to continuously collect and transmit real-time agricultural data. The integration of wireless communication technologies such as LoRa and cloud platforms enables farmers to monitor field conditions remotely and make informed decisions regarding irrigation

and crop management. Therefore, there is a need for an IoT-enabled smart sugarcane farming system that can provide real-time monitoring, efficient water utilization, and improved crop productivity while reducing manual effort.



Figure 1.1 Sugarcane field

Since agricultural fields may not always have direct power availability, low-power battery-operated sensor nodes are essential for reliable field deployment. Efficient power management circuits and voltage regulation are necessary to ensure stable operation of microcontrollers and sensors in outdoor environments.

1.2 Motivation Behind the Project

The primary motivation behind developing this project arises from the increasing challenges faced in conventional agricultural practices. Water scarcity, inefficient irrigation methods, unpredictable weather conditions, and a lack of continuous field monitoring are some of the major issues affecting agricultural productivity. Sugarcane cultivation requires a significant amount of water throughout different stages of plant growth, making efficient water management extremely important.

In traditional farming systems, irrigation decisions are usually made based on visual observation or fixed schedules without considering actual soil conditions. Such approaches may result in either under-irrigation or over-irrigation, both of which negatively affect crop yield and resource utilization. Moreover, frequent field visits by farmers become difficult, especially for large agricultural lands, increasing labor requirements and reducing efficiency.

Recent advancements in IoT technologies, wireless communication systems, and cloud-based monitoring solutions have created opportunities for implementing intelligent farming techniques. Low-power long-range communication technologies such as LoRa allow sensor data to be transmitted over large distances with minimal power consumption, making them suitable for agricultural applications. The availability of cloud platforms and web dashboards further enables remote monitoring and historical data analysis.

This project was motivated by the need to develop an intelligent monitoring system that can provide farmers with continuous information about field conditions and assist them in making better irrigation decisions. Instead of implementing a completely automated irrigation system, this project focuses on helping farmers by providing real-time sensor data and allowing irrigation control through a mobile application based on actual field requirements.

1.3 Introduction to IoT Enabled Smart Sugarcane Farming System

IoT Enabled Smart Sugarcane Farming Systems are intelligent agricultural monitoring and irrigation support systems that utilize sensors, wireless communication technologies, cloud platforms, and mobile applications to improve farming efficiency and resource management. These systems continuously monitor important environmental and soil parameters from different regions of agricultural fields and provide real-time information to farmers for effective crop management.

A typical smart sugarcane farming system consists of multiple sensor nodes and a central gateway unit. Sensor nodes are generally developed using microcontrollers integrated with various environmental sensors such as temperature and humidity sensors, soil moisture sensors, and rain sensors. These sensors continuously collect field data from different locations

of the agricultural land. Since agricultural fields may not always have direct power availability, sensor nodes are commonly powered using rechargeable batteries along with voltage regulation circuits to provide stable power supply for controllers and sensors.

For long-range wireless communication between sensor nodes and the gateway, technologies such as LoRa are widely used. LoRa communication offers advantages including long communication range, low power consumption, and reliable wireless transmission, making it suitable for agricultural environments. The collected sensor data from multiple nodes is transmitted wirelessly to a central gateway for processing and monitoring.

The gateway unit acts as the central processing and communication component of the system. It receives sensor information from all deployed nodes, processes the collected data, and may display the information locally using display modules such as LCDs. The gateway is also commonly connected to the internet through Wi-Fi or other communication technologies, enabling remote monitoring and cloud-based data transmission.

Smart farming systems may also include irrigation support mechanisms consisting of water pumps, relay modules, and solenoid valves for controlling water flow to different agricultural sections. Depending on system design, irrigation can be controlled either automatically based on sensor conditions or manually through mobile applications and remote interfaces. This helps farmers manage irrigation efficiently according to crop requirements and environmental conditions.

For remote monitoring and data storage, cloud platforms and web-based dashboards are commonly integrated into IoT agricultural systems. These platforms store real-time and historical sensor information, provide graphical visualization of field conditions, and generate alerts or recommendations whenever parameters exceed predefined thresholds. Such features assist farmers in making informed decisions related to irrigation, crop monitoring, and resource utilization.

Overall, IoT Enabled Smart Sugarcane Farming Systems combine real-time sensing, wireless communication, cloud computing, and remote accessibility technologies to improve agricultural productivity, optimize water usage, reduce manual effort, and support sustainable farming practices.

1.4 Applications

The IoT Enabled Smart Sugarcane Farming System can be applied in various agricultural and environmental monitoring applications. One of the primary applications of the system is in sugarcane plantations where continuous monitoring of field conditions is necessary for effective crop management. By providing real-time information regarding soil moisture, temperature, humidity, and rainfall, the system helps farmers optimize irrigation practices and improve crop productivity.

The system can also be used in precision agriculture, where farming decisions are based on real-time data collected from sensors. Precision agriculture helps minimize resource wastage and enables efficient utilization of water, fertilizers, and labor. Since the proposed system uses wireless communication and cloud technologies, it can also support large-scale agricultural land monitoring where manual inspection becomes difficult.

Apart from sugarcane farming, the system can be adapted for other crops requiring environmental monitoring and irrigation management. It can be used in greenhouse monitoring systems, agricultural research activities, environmental condition monitoring, and smart irrigation systems. Furthermore, the cloud-based monitoring and data storage features make the system suitable for long-term crop analysis and research purposes.

1.5 Limitations

The performance of the system depends on internet connectivity because Wi-Fi communication is required for transmitting data to the cloud platform and enabling remote monitoring features. In areas with poor network availability, remote monitoring may not function effectively.

As the number of deployed nodes increases, system complexity and maintenance requirements also increase. Additional nodes require more hardware components, increased power consumption, and proper network management. Furthermore, the initial installation cost of sensors, communication modules, and cloud infrastructure may be relatively high for small-scale farmers.

Despite these limitations, the proposed system provides a practical and cost-effective solution for improving agricultural monitoring and supporting informed irrigation management in sugarcane cultivation.

CHAPTER - 2

BACKGROUND THEORY

This chapter presents the theoretical foundation of the proposed IoT Enabled Smart Sugarcane Farming System by discussing existing agricultural monitoring systems and their limitations. It explains the proposed smart farming solution by describing the major hardware and software components and their functions. The chapter also discusses the operating principles of the system, including real-time sensing, wireless communication, cloud integration, remote monitoring, and irrigation support mechanisms.

2.1 Discussion of background work related to the project

Recent developments in digital communication and Internet of Things (IoT) technologies have significantly transformed conventional agricultural practices. Smart agriculture combines sensors, wireless communication protocols, cloud computing, and remote monitoring technologies to improve farming efficiency and crop productivity. These technologies form the basis of the proposed IoT Enabled Smart Sugarcane Farming System, which is designed to continuously monitor environmental conditions and support farmers in making informed irrigation decisions.

The integration of IoT technology in agriculture enables real-time monitoring of field conditions and minimizes the dependence on manual observation. Advanced communication technologies such as LoRa provide long-range, low-power wireless communication suitable for agricultural environments. Cloud platforms further allow data storage, trend analysis, and remote access to field information from any location. In battery-powered agricultural monitoring systems, efficient power management is important for ensuring reliable long-term operation. Voltage regulation circuits such as buck converters and linear regulators are commonly used to provide stable supply voltages for microcontrollers and sensors.

Modern agricultural systems are moving from traditional manual monitoring toward intelligent and data-driven approaches. Smart farming technologies help optimize water usage, reduce labor requirements, improve crop health monitoring, and increase overall productivity. By

combining sensor networks with cloud connectivity, agricultural systems can provide timely alerts and recommendations to farmers.

2.1.1 Overview of Smart Agricultural Monitoring Systems

Smart agricultural monitoring systems are designed to continuously observe field conditions and assist farmers in crop management activities. Conventional agricultural systems generally depend on manual field inspections and fixed irrigation schedules. These methods may not provide accurate information about actual soil and environmental conditions and often result in inefficient resource utilization.

Modern smart farming systems use various sensors to monitor parameters such as soil moisture, temperature, humidity, and rainfall. Sensor data is transmitted through communication technologies to central systems where the information can be analyzed and displayed. These systems help farmers remotely monitor field conditions and improve decision-making regarding irrigation and crop management.

Recent trends in agricultural technology have shifted from basic monitoring systems toward intelligent and adaptive systems capable of providing recommendations and predictive analysis. Real-time monitoring combined with cloud-based storage allows historical data analysis and trend visualization. Such systems improve efficiency while minimizing unnecessary water usage and operational costs.

Furthermore, emerging technologies such as artificial intelligence, machine learning, and edge computing are creating new possibilities for future agricultural systems. These technologies can analyze historical data, predict environmental conditions, and provide automated recommendations for irrigation and crop management.

2.1.2 Limitations of Existing Systems

Despite the availability of several agricultural monitoring systems, many limitations still exist:

- **Lack of Continuous Monitoring:** Many traditional farming systems rely on periodic manual observation and cannot provide continuous real-time information.
- **Limited Communication Range:** Some wireless technologies provide limited communication distance and are not suitable for large agricultural fields.
- **High Power Consumption:** Certain communication systems consume high amounts of power, reducing operational efficiency.
- **Limited Remote Accessibility:** Many systems do not support cloud integration and remote monitoring capabilities.
- **High Cost of Fully Automated Systems:** Advanced smart irrigation systems can involve high installation and maintenance costs.
- **Limited Decision Support:** Several systems only provide sensor readings without offering alerts or recommendations.

2.1.3 Proposed IoT Enabled Smart Sugarcane Farming System

To overcome the limitations of existing agricultural systems, the proposed system utilizes multiple sensor nodes, wireless communication technology, cloud integration, and a mobile-based monitoring platform.

Microcontroller Units (MCUs)

- **STM32 Nucleo F401RE:** Acts as the controller for individual sensor nodes. It processes sensor inputs and manages LoRa communication.
- **ESP32 Gateway:** Functions as the central gateway that receives data from all sensor nodes and transmits it to the cloud through Wi-Fi.

Sensors and Monitoring Components

- **DHT11 Sensor:** Measures environmental temperature and humidity around the sugarcane field.
- **Capacitive Soil Moisture Sensor:** Measures soil moisture levels and helps determine irrigation requirements.
- **Rain Sensor:** Detects rainfall conditions and prevents unnecessary irrigation.

- **Power Management Components:**

1. **2 × 3.7V, 2400mAh Li-ion Batteries:** Provide portable power supply for each STM32 sensor node deployed in the agricultural field.
2. **Buck Converter Module:** Regulates the battery voltage and provides a stable 5V supply for the STM32 microcontroller and LoRa module.
3. **AMS1117 Voltage Regulator:** Provides a stable 3.3V output for powering the DHT11 sensor, soil moisture sensor, and rain sensor.

Communication Modules

- **SX1278 LoRa Module:** Provides long-range wireless communication between sensor nodes and the gateway with low power consumption.
- **Wi-Fi Module in ESP32:** Enables internet connectivity for remote monitoring and cloud data transmission.
- To avoid LoRa packet collision between multiple sensor nodes, a time-staggered communication approach is implemented. Each node operates with different RTC wake-up intervals and initial transmission offsets, ensuring that sensor nodes transmit data at different times. This improves communication reliability and prevents simultaneous LoRa packet transmission to the ESP32 gateway.

Irrigation Components

- **5V Solenoid Valves:** Control water flow for irrigation in different field sections.
- **5V Water Pump:** Supplies water for irrigation purposes.
- **4-Channel Relay Module:** Interfaces and controls the pump and solenoid valves.

Software and Cloud Integration

- **Blynk Mobile Application:** Allows farmers to manually control irrigation based on field conditions.
- **Web Dashboard:** Displays real-time sensor data and historical trends.
- **Supabase Database:** Stores field data securely in the cloud for analysis and monitoring.

- **Alert Mechanism:** Provides recommendations and alerts when parameters exceed optimal thresholds.

2.1.4 Working Principles

- **Data Acquisition and Monitoring:** Sensor nodes continuously measure field conditions including temperature, humidity, soil moisture, and rainfall. Each STM32 node is powered using two 3.7V Li-ion batteries. A buck converter provides a stable 5V supply to the STM32 controller, while an AMS1117 voltage regulator provides regulated 3.3V power for the sensors. Data acquisition occurs periodically at different locations within the sugarcane field.
- **Data Transmission through LoRa Communication:** The STM32 sensor nodes process collected sensor data and transmit the readings wirelessly using SX1278 LoRa modules. LoRa technology enables long-distance communication while consuming minimal power, making it suitable for battery-operated agricultural sensor nodes.
- **Gateway Processing and Display:** The ESP32 gateway receives information from all three sensor nodes. The received data is displayed sequentially on an I2C LCD display for local monitoring.
- **Remote Monitoring and Cloud Communication:** The ESP32 forwards the received data through Wi-Fi to a remote server. The web dashboard retrieves this information and displays live sensor readings from individual nodes. The dashboard also stores information in the Supabase cloud database for historical analysis.
- **Irrigation Control Mechanism:** Farmers monitor soil moisture data and field conditions through the dashboard or Blynk application. Based on the water requirements of sugarcane during different growth stages, the farmer manually controls the corresponding solenoid valves and water pump for irrigation.

- **Smart Alerts and Recommendations:** When field parameters exceed predefined threshold values, the dashboard generates alerts and precautionary recommendations.

Examples include:

1. Low soil moisture → Irrigate the field
2. High temperature → Increase irrigation or provide shade
3. Rain detected → Pause irrigation to avoid waterlogging

- **Integration and Testing**

1. **Individual Component Testing:** Each sensor, communication module, relay, and actuator is tested independently to verify proper operation.
2. **Communication Testing:** LoRa transmission between sensor nodes and the ESP32 gateway is tested for reliability and communication range.
3. **System Integration Testing:** All hardware and software components are integrated and tested together.
4. **Field Testing:** The complete system is deployed and tested under real agricultural conditions to validate performance and monitoring accuracy.

The proposed IoT Enabled Smart Sugarcane Farming System provides an efficient and scalable approach for agricultural monitoring by combining real-time sensing, wireless communication, cloud storage, and remote accessibility. The system supports farmers in making informed irrigation decisions and contributes to improved water management and agricultural productivity.

2.1.5 Power Management in Sensor Nodes

To improve battery life and reduce power consumption, the STM32 sensor nodes operate in low-power STOP mode after completing sensing and LoRa data transmission. The RTC wake-up timer periodically wakes the microcontroller for sensor acquisition and communication tasks. After transmission, the node re-enters STOP mode, significantly reducing power consumption and supporting long-duration battery-powered agricultural deployment.

CHAPTER - 3

LITERATURE REVIEW

This chapter reviews existing research related to IoT-based smart agriculture systems and modern farming technologies. It discusses the limitations of existing monitoring and irrigation approaches and highlights the improvements introduced in the proposed system. The review supports the selected architecture and emphasizes the need for smart farming solutions.

3.1 Review of literature

[1] Soumil Heble et al. (2018), “A Low Power IoT Network for Smart Agriculture”, Indian Institute of Technology Hyderabad.

This paper proposes a low-cost and low-power IoT network architecture for smart agriculture applications. The system uses wireless sensor networks for monitoring environmental and soil parameters such as soil moisture, temperature, humidity, and light intensity. The proposed architecture focuses on reducing power consumption and increasing network lifetime while maintaining reliable communication. The study demonstrated improved energy efficiency and cost-effectiveness compared to existing agricultural monitoring systems, making it suitable for large-scale agricultural deployment.

[2] Pavankumar Naik et al. (2018), “Automation of Irrigation System using IoT”, International Journal of Engineering and Manufacturing Science, Vol. 8, No. 1, pp. 77–88.

This paper presents an IoT-based automated irrigation system designed to reduce water wastage and improve irrigation efficiency in agriculture. The system uses soil moisture, temperature, and humidity sensors connected to an Arduino microcontroller to monitor field conditions continuously. Based on soil moisture values, the irrigation motor is automatically controlled through relay switching. Sensor readings and motor status are displayed on an Android application, enabling remote monitoring and efficient water management. The system demonstrates the practical application of IoT technology in modern smart irrigation systems.

[3] Kaveri S. Kamble et al. (2019), “A Review on IoT Based Smart Agriculture for Sugarcane”, International Journal of Engineering Development and Research (IJEDR), Vol. 7, Issue 2.

This review paper discusses IoT-based smart agriculture techniques specifically developed for sugarcane farming. The proposed system monitors real-time soil and environmental parameters using sensors such as soil moisture, soil temperature, humidity, and pH sensors. The collected data is periodically transmitted to agro-cloud storage using IoT technology for analysis and decision-making. The study highlights the role of cloud platforms, wireless sensor networks, and IoT technologies in improving sugarcane productivity, irrigation management, and fertilizer optimization.

[4] E. Wang et al. (2019), “Smarter Irrigation Scheduling in the Sugarcane Farming System using the Internet of Things”, Proceedings of the 41st Annual Conference of the Australian Society of Sugar Cane Technologists.

This paper focuses on improving irrigation scheduling in sugarcane farming using IoT technologies and intelligent decision-support systems. The study integrates automated irrigation systems with IrrigWeb software to optimize irrigation scheduling based on crop conditions, weather information, and soil moisture data. The developed system reduces manual effort by automatically uploading irrigation and rainfall data and generating efficient irrigation schedules. The proposed solution demonstrates how IoT technology can improve water-use efficiency and support precision agriculture in sugarcane farming.

[5] E. Wang et al. (2020), “Development of a Closed-Loop Irrigation System for Sugarcane Farms using the Internet of Things”, Computers and Electronics in Agriculture, Vol. 172.

This paper presents the development of a closed-loop IoT-based irrigation management system for sugarcane farms. The system integrates automated irrigation hardware, wireless communication technologies, sensors, and irrigation decision-support tools to improve water-use efficiency and crop productivity. The developed system automatically calculates irrigation schedules by considering weather conditions, soil properties, crop requirements, and pumping constraints. The study demonstrates that IoT-enabled irrigation systems can reduce water wastage, improve agricultural sustainability, and support efficient smart farming practices.

[6] Biruk Abate Demisse et al. (2021), *“Improving the Efficiency of Sugarcane Production using Internet of Things (IoT): The Case of Wonji/Shoa Sugar Factory”*, St. Mary’s University, Ethiopia.

This paper presents an IoT-based smart sugarcane monitoring system developed to improve sugarcane productivity and irrigation management. The system utilizes sensors such as soil moisture, temperature, and humidity sensors integrated with Arduino microcontrollers and GSM/GPRS communication modules for real-time field monitoring. Sensor data is transmitted to the ThingSpeak cloud platform for storage, graphical visualization, and analysis. The proposed system helps agronomists and farmers make informed decisions regarding irrigation and environmental management, thereby improving sugarcane production efficiency.

[7] Pengwen Wang et al. (2021), *“An Improved Multilayer Perceptron Approach for Detecting Sugarcane Yield Production in IoT Based Smart Agriculture”*, *Microprocessors and Microsystems*, Vol. 82, 103822.

This paper presents an IoT-based smart agriculture system that uses artificial neural networks and machine learning algorithms for predicting sugarcane yield production. The proposed system utilizes IoT technologies and environmental data collection to improve precision agriculture and optimize sugar production processes. An improved Multilayer Perceptron (MLP) algorithm was developed to predict sugar yield with high accuracy using parameters related to sugarcane cultivation and production. Experimental results demonstrated high prediction accuracy and reduced error rates, highlighting the potential of machine learning and IoT technologies in intelligent agricultural management and crop yield prediction.

[8] Nqobile Shika et al. (2023), *“Investigating Barriers to Adoption of Internet of Things (IoT) in Agriculture: A Case of Smallholder Sugarcane Farmers in Ndwedwe Local Municipality”*, University of Johannesburg.

This research investigates the challenges faced by smallholder sugarcane farmers in adopting IoT technologies in agriculture. The study identifies major barriers such as high implementation costs, poor infrastructure, limited digital literacy, security concerns, and lack of technical support. The paper highlights that despite the benefits of smart agriculture technologies, many farmers still rely on traditional farming practices due to financial and

technical limitations. The study emphasizes the need for affordable, user-friendly, and scalable IoT solutions for agricultural applications.

[9] Simone Coniglione et al. (2023), “*Design of IoT Node for Smart Agriculture*”, Master’s Degree Thesis, Politecnico di Torino, Italy.

This thesis presents the design and development of a long-range low-power IoT hardware platform for smart agriculture applications. The system is based on the STM32WL55CC microcontroller and supports LPWAN communication protocols such as LoRa, GFSK, and BPSK for reliable long-distance communication in agricultural environments. The proposed design includes power supply management, RF communication modules, PCB design, firmware configuration, and LoRa communication testing. The developed platform simplifies the implementation of IoT-based agricultural monitoring systems and improves wireless communication reliability for precision agriculture applications.

[10] Abdellatif Soussi et al. (2024), “*Smart Sensors and Smart Data for Precision Agriculture: A Review*”, *Sensors* 2024, Vol. 24, 2647.

This review paper discusses the role of smart sensors, IoT technologies, wireless sensor networks, cloud computing, artificial intelligence, and data analytics in precision agriculture. The study highlights how smart agricultural systems utilize environmental sensors for monitoring parameters such as soil moisture, temperature, humidity, CO₂ levels, and light intensity to optimize farming operations. The paper also reviews different IoT platforms, communication protocols, and intelligent algorithms used in modern agriculture for irrigation management, disease detection, and resource optimization. The research emphasizes the importance of integrating smart sensors and data-driven technologies for sustainable and efficient farming practices.

[11] D. T. Huong et al. (2024), “*Enhancing Precision Agriculture: An IoT-based Smart Monitoring System Integrated LoRaWAN, ML and AR*”, *EAI Endorsed Transactions on Smart Cities*.

This paper presents an IoT-based smart agriculture monitoring system integrated with LoRaWAN communication, Machine Learning (ML), and Augmented Reality (AR)

technologies. The proposed system collects environmental data such as soil moisture, air humidity, and temperature using IoT sensor nodes and transmits the information using LoRaWAN for long-range communication. Machine learning algorithms are used for predicting environmental conditions and crop-related parameters, while AR-based interfaces provide real-time visualization and farm monitoring. The system improves agricultural productivity, supports predictive analysis, and enables efficient farm management through real-time monitoring and intelligent data processing.

[12] Ying Zhang et al. (2025), “Research and Development of an IoT Smart Irrigation System for Farmland Based on LoRa and Edge Computing”, *Agronomy* 2025, Vol. 15, 366.

This paper proposes an IoT-based smart irrigation system using LoRa communication and edge computing technologies for precision irrigation management. The system uses STM32 microcontrollers, LoRa modules, GPS modules, and environmental sensors to collect soil moisture and weather data from farmland. An edge-computing-based irrigation decision model was developed to optimize water usage and improve irrigation efficiency. The system also includes a mobile application for remote monitoring and precise control of irrigation valves. Experimental results demonstrated reliable long-range communication up to 4 km and effective water-saving irrigation performance.

[13] Gustavo Gimenes et al. (2025), “Low-Cost Remote Sensing Module for Agriculture 4.0 Based on STM32”, *Engineering Proceedings* 2025, Vol. 118, 97.

This paper presents the development of a low-cost remote sensing module for Agriculture 4.0 applications using the STM32F411CEU6 microcontroller. The proposed system monitors environmental parameters such as soil moisture, temperature, humidity, and atmospheric pressure using multiple sensors interfaced through I2C, SPI, and ADC communication protocols. Wireless communication is established using nRF24L01 modules in a mesh network architecture. The system is powered using solar cells and Li-Po batteries, making it suitable for remote agricultural environments. The developed solution aims to provide affordable smart agriculture technologies for small and medium-scale farmers.

[14] Mohammed Almutairi et al. (2025), “*IoT–Cloud Integration Security: A Survey of Challenges, Solutions, and Directions*”, *Electronics* 2025, Vol. 14, 1394.

This survey paper discusses the integration of IoT systems with cloud computing and the associated security challenges in IoT-cloud ecosystems. The study highlights how IoT devices generate large amounts of real-time data that are processed and stored using cloud platforms for monitoring and analytics. The paper reviews security vulnerabilities such as data breaches, API attacks, virtualization threats, and privacy issues in IoT-cloud systems. It also discusses mitigation techniques including encryption, lightweight cryptography, blockchain-based frameworks, and AI-driven anomaly detection. The research emphasizes the importance of secure and scalable IoT-cloud integration for smart applications such as agriculture and industrial automation.

3.2 Summary of literature

- IoT-based smart agriculture systems play an important role in improving irrigation management, crop monitoring, and agricultural productivity.
- Several researchers developed systems using sensors, wireless communication technologies, cloud platforms, and automation techniques for agricultural monitoring.
- Environmental parameters such as soil moisture, temperature, humidity, and rainfall were commonly monitored using sensor-based systems.
- LoRa communication technology was widely adopted due to its long communication range and low power consumption suitable for agricultural environments.
- Cloud integration enabled real-time remote monitoring, historical data storage, and graphical visualization of sensor information.
- Some studies incorporated machine learning and intelligent decision-making techniques for improving irrigation scheduling and crop management.
- Precision irrigation techniques helped reduce water wastage and improve farming efficiency.
- Existing systems still faced limitations such as:
 - High implementation cost
 - Limited communication range

- High power consumption
- Lack of scalability
- Insufficient remote accessibility
- Certain systems mainly focused on automatic irrigation without providing real-time cloud monitoring and smart recommendation features.
- The observations from the literature motivated the development of an IoT Enabled Smart Sugarcane Farming System combining:
 - Real-time sensing
 - LoRa communication
 - Cloud-based monitoring
 - Remote irrigation support
 - Low-power sensor node operation
- The proposed approach aims to provide an efficient, practical, and scalable smart agriculture solution for improved irrigation management and agricultural productivity.

CHAPTER - 4

PROBLEM STATEMENT

This chapter describes the problems associated with traditional sugarcane farming and the need for efficient monitoring and irrigation management. It presents the objectives of the proposed system and explains the methodology adopted for implementing the IoT Enabled Smart Sugarcane Farming System.

4.1 Problem statement

Agriculture plays an essential role in sustaining economic growth and food production, and efficient crop management is necessary for achieving higher productivity. Sugarcane cultivation requires continuous monitoring of environmental and soil parameters such as temperature, humidity, soil moisture, and rainfall. Traditional farming methods primarily rely on manual inspection and fixed irrigation schedules, which often lead to inefficient use of water resources and reduced crop yield. Farmers need to physically visit fields to observe crop conditions, which becomes difficult and time-consuming, especially in large agricultural areas.

Improper irrigation management can result in either over-irrigation or under-irrigation. Excessive irrigation causes water wastage and waterlogging, while insufficient irrigation affects crop growth and quality. In addition, changing environmental conditions and unpredictable weather patterns further increase the difficulty of maintaining proper field conditions. Existing agricultural systems often lack continuous monitoring, long-range communication capability, remote accessibility, and intelligent decision support mechanisms.

To address these limitations, the proposed project aims to develop an IoT Enabled Smart Sugarcane Farming System using sensor networks, LoRa communication, cloud technology, and remote monitoring platforms. The system continuously monitors field conditions and assists farmers in making informed irrigation decisions, thereby improving water management and agricultural productivity.

4.2 Objectives

To address the need for efficient agricultural monitoring and irrigation management, the project focuses on developing an intelligent and connected farming system. The key objectives are:

- To review literature on IoT based smart farming systems, LoRa wireless communication, cloud monitoring platforms and irrigation control systems for agriculture applications.
- To arrive at the requirements and design specifications for IoT enabled Smart Sugarcane Farming System based on the literature review.
- To develop functional block diagram of Smart Sugarcane Farming System.
- To develop three individual sensor nodes using STM32 Nucleo F401RE integrated with sensors and develop the ESP32 gateway node.
- To develop a Smart Sugarcane Farming System for real-time monitoring and remote irrigation control using IoT and cloud technology.
- To test and validate the implemented “Smart Sugarcane Farming System” under real agricultural field conditions, for its working functionality.

4.3 Methodology

The development of the IoT Enabled Smart Sugarcane Farming System followed a systematic engineering approach to ensure successful implementation and alignment with the project objectives. The methodology adopted for the development and implementation of the system is explained in the following phases:

1. Literature Review

A detailed study of existing IoT-based smart agriculture systems, wireless sensor networks, irrigation management systems, cloud monitoring platforms, and LoRa communication technologies was carried out. Various research papers, journals, and technical articles related to precision agriculture and smart irrigation were reviewed. The limitations of conventional

farming methods and existing smart farming systems were analyzed to identify the need for an efficient, low-power, and remotely accessible agricultural monitoring system.

2. Requirement Analysis and Objective Definition

Based on the literature survey and field requirements in sugarcane cultivation, the system requirements were identified. Important agricultural parameters such as soil moisture, temperature, humidity, and rainfall were selected for continuous monitoring. The project objectives were defined to support real-time environmental monitoring, efficient irrigation management, cloud-based remote access, and low-power operation suitable for field deployment.

3. System Design and Specification

A complete system architecture was designed consisting of distributed STM32-based sensor nodes, LoRa wireless communication modules, an ESP32 gateway, cloud database integration, irrigation control units, and monitoring interfaces. Hardware and software specifications were finalized after considering communication range, power consumption, reliability, and cost effectiveness for agricultural applications.

4. Functional Block Diagram Development

A functional block diagram of the proposed system was developed to represent the interaction between sensors, microcontrollers, communication modules, cloud platforms, and irrigation components. The block diagram illustrated the flow of sensor data from field nodes to the ESP32 gateway and subsequently to the cloud database and monitoring dashboard.

5. Component Selection and Hardware Integration

Suitable hardware components were selected and integrated based on project requirements. STM32 Nucleo F401RE boards were used for sensor nodes, while ESP32 was selected as the gateway controller due to its built-in Wi-Fi capability. Environmental monitoring was performed using DHT11 sensors, capacitive soil moisture sensors, and rain sensors. SX1278 LoRa modules were used for long-range wireless communication. Relay modules, solenoid valves, and a water pump were integrated for irrigation control. Li-ion batteries, buck

converters, and AMS1117 voltage regulators were used for portable and regulated power supply operation.

6. Sensor Calibration and Initial Testing

All sensors were individually tested and calibrated under different environmental conditions to ensure proper operation and accurate readings. Soil moisture sensors were tested in dry and wet soil conditions, while DHT11 sensors were verified for temperature and humidity measurements. Rain sensors were tested by simulating rainfall conditions. Sensor outputs were validated using serial monitor readings before complete system integration.

7. Development of STM32 Sensor Node Firmware

Embedded firmware for the STM32 sensor nodes was developed using STM32CubeIDE. The firmware was designed to perform sensor data acquisition, ADC conversion, GPIO control, RTC wake-up scheduling, and LoRa data transmission. Low-power STOP mode functionality was implemented to reduce battery consumption during idle periods. The RTC wake-up timer periodically activated the controller for sensing and communication operations.

8. LoRa Communication Configuration

SX1278 LoRa modules were configured with suitable parameters such as operating frequency, spreading factor, bandwidth, and transmission intervals for reliable long-range communication. LoRa communication testing was performed between STM32 sensor nodes and the ESP32 gateway to verify communication reliability and transmission range under field conditions. To avoid packet collision between multiple sensor nodes, a time-staggered transmission approach was implemented. Different RTC wake-up intervals and initial transmission offsets were configured for each node so that data packets were transmitted at different times.

9. ESP32 Gateway Development

Firmware for the ESP32 gateway was developed using the Arduino IDE. The ESP32 gateway was programmed to receive LoRa packets from multiple sensor nodes, process the received sensor information, and display the data on an I2C LCD module. The gateway also established

Wi-Fi connectivity and transmitted sensor information to the cloud database through HTTP requests and JSON formatting.

10. Cloud Database and Backend Configuration

A cloud-based monitoring infrastructure was developed using Supabase and backend server technologies. REST APIs were configured for receiving sensor data from the ESP32 gateway. Sensor readings, node identification information, timestamps, and logs were stored securely in the cloud database for real-time monitoring and historical analysis.

11. Web Dashboard and Mobile Application Development

A web-based dashboard was developed for displaying real-time sensor readings, node-wise monitoring, graphical visualization of historical data, and alert notifications. The dashboard provided farmers with remote access to field conditions through internet connectivity. The Blynk mobile application was configured for irrigation control. Farmers could remotely control the water pump and solenoid valves using the application based on real-time field conditions and crop water requirements.

12. Irrigation System Integration

Relay modules were interfaced with the ESP32 gateway to control the operation of the water pump and solenoid valves. Irrigation testing was performed by activating individual valves and pump control through the Blynk application. Proper water flow control and relay switching operations were verified under practical conditions.

13. Power Management Implementation

Efficient power management techniques were implemented to support battery-powered field deployment. Two 3.7V Li-ion batteries were used for each sensor node. Buck converter modules provided regulated 5V supply to the STM32 controller and LoRa module, while AMS1117 voltage regulators generated stable 3.3V supply for the sensors. Low-power STOP mode and RTC wake-up scheduling were implemented in STM32 sensor nodes to significantly reduce power consumption and improve battery life for long-duration agricultural operation.

14. System Integration and Field Deployment

After individual hardware and software testing, all modules were integrated into a complete IoT-enabled smart farming system. The integrated system consisting of sensor nodes, ESP32 gateway, cloud database, web dashboard, and irrigation control setup was deployed in a real sugarcane field at VC Farms, Mandya for practical testing and performance evaluation.

15. Testing, Validation, and Performance Analysis

The implemented system was tested under real agricultural conditions to evaluate its overall performance. Sensor accuracy, LoRa communication reliability, cloud connectivity, dashboard response, irrigation control functionality, and low-power operation were analyzed. The obtained results were validated through continuous monitoring and practical field observations to ensure reliable operation of the proposed smart sugarcane farming system.

CHAPTER - 5

SYSTEM DESIGN OF “IoT Enabled Smart Sugarcane Farming System”

This chapter describes the overall system architecture of the proposed IoT Enabled Smart Sugarcane Farming System. It explains the functional organization of various hardware and software components involved in the system. The chapter also presents the block diagram and describes the operation of individual modules used for sensing, communication, remote monitoring, and irrigation management.

5.1 Block Diagram of the Proposed IoT Enabled Smart Sugarcane Farming System

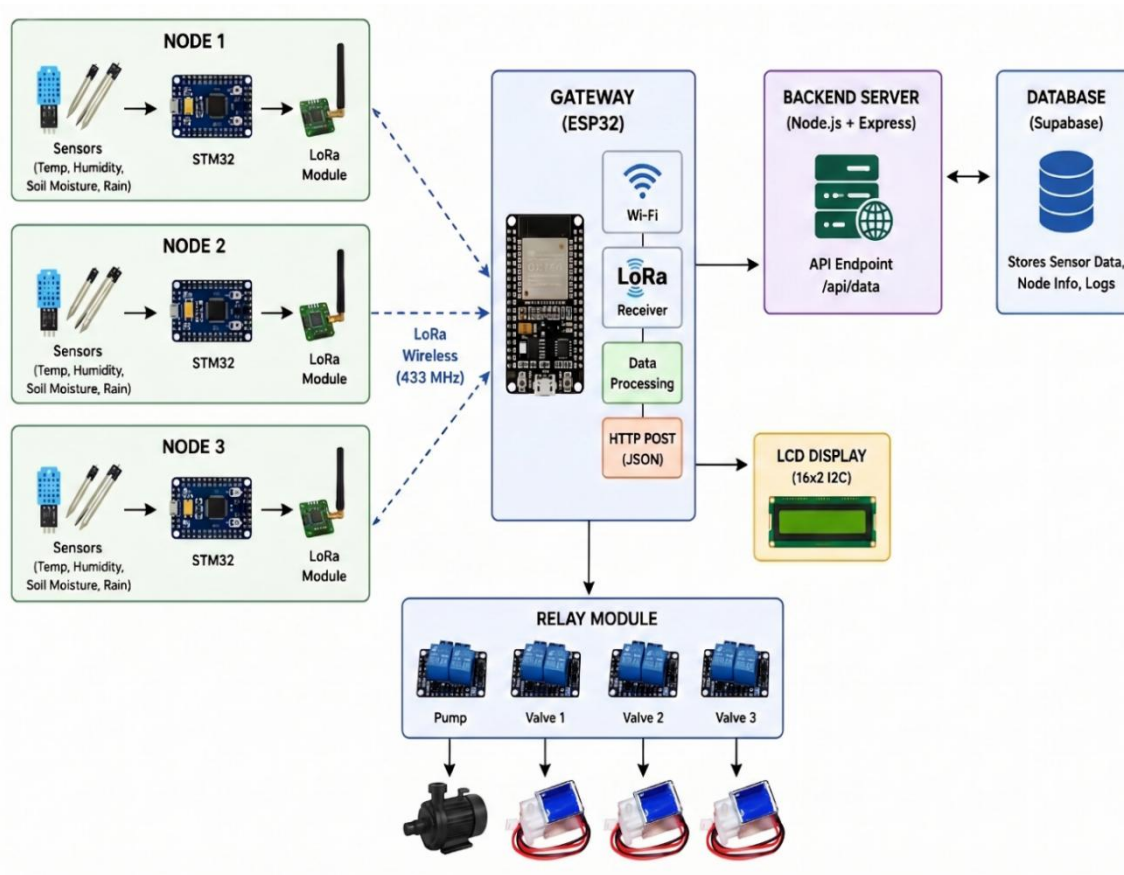


Figure 5.1 : Block diagram of the design

5.2 Description of Block Diagram

The proposed system architecture shown in Figure 5.1 consists of three sensor nodes deployed at different locations in the sugarcane field and one central ESP32 gateway unit. Each sensor node contains environmental sensors including a DHT11 sensor for temperature and humidity measurement, a capacitive soil moisture sensor for determining soil moisture level, and a rain sensor for rainfall detection. The sensor data collected by each node is processed by an STM32 Nucleo F401RE microcontroller.

The processed data is transmitted wirelessly through SX1278 LoRa modules operating at 433 MHz frequency. LoRa communication provides long-range and low-power transmission suitable for agricultural applications. The ESP32 gateway receives sensor information from all three nodes and processes the incoming data.

To improve communication reliability, the sensor nodes use staggered RTC wake-up intervals for LoRa transmission. This prevents simultaneous packet transmission and minimizes communication collision between multiple nodes.

The gateway displays sensor readings sequentially on an I2C-based LCD display for local monitoring. It also establishes Wi-Fi communication and transmits the received data to a backend server through HTTP POST requests in JSON format. The backend server communicates with the Supabase cloud database where sensor information, node details, and logs are stored.

For irrigation support, the ESP32 gateway is connected to a relay module that controls three solenoid valves and a water pump. Through the Blynk application, farmers can manually operate irrigation based on field conditions and moisture requirements of sugarcane during different growth stages.

5.3 Explanation of individual blocks

1. Sensor Nodes (Node 1, Node 2, Node 3)

The sensor unit consists of a DHT11 sensor, capacitive soil moisture sensor, and rain sensor used shown in Figures 5.2, 5.3 and 5.4 for monitoring environmental and soil conditions in the sugarcane field. The DHT11 sensor measures temperature and humidity, the capacitive soil moisture sensor measures soil water content based on capacitance variation, and the rain sensor detects rainfall conditions through conductivity changes on the sensing surface. The STM32 microcontroller collects and processes the sensor data and transmits the information wirelessly to the ESP32 gateway using LoRa communication.

Functions:

- Temperature and humidity monitoring
- Soil moisture measurement
- Rainfall detection
- Real-time environmental monitoring

Advantages:

- Accurate soil moisture sensing
- Low power operation
- Reliable wireless data transmission

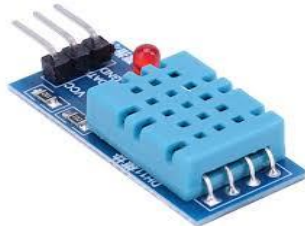


Figure 5.2: DHT11 sensor

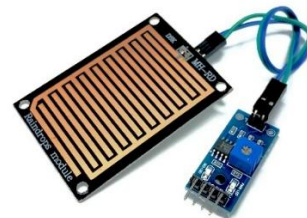


Figure 5.3: Rain sensor



Figure 5.4: Capacitive soil moisture sensor

2. STM32 Microcontroller

The STM32 Nucleo F401RE shown in Figure 5.5 acts as the main processing and control unit of each sensor node. It interfaces with all sensors, processes acquired data, controls LoRa communication, and manages low-power operation. The STM32 microcontroller was selected due to its high processing capability, low power consumption, multiple communication interfaces, and suitability for embedded IoT applications.

The controller operates in low-power STOP mode during idle periods to conserve battery energy. It periodically wakes up using RTC interrupts, performs sensing and communication tasks, and returns to sleep mode.

Functions:

- Sensor interfacing and data acquisition
- Data processing and formatting
- LoRa communication management
- RTC-based periodic wake-up operation
- Low-power energy-efficient operation
- Control of sensor node activities

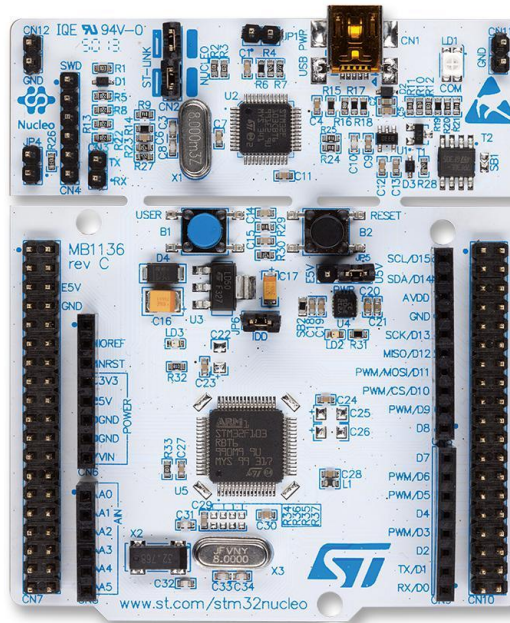


Figure 5.5: STM32 Nucleo – F401RE

3. Li-ion Battery Power Supply

Each sensor node is powered using two 3.7V, 2400mAh Li-ion rechargeable batteries as shown in Figure 5.6, connected to provide reliable portable power for field deployment. Since agricultural fields may not always have direct electrical power availability, battery operation enables independent and flexible deployment of sensor nodes.

Functions:

- Provides portable power supply
- Supports long-duration operation
- Enables remote field deployment
- Supplies power to controllers, sensors, and communication modules

Advantages:

- Rechargeable and reusable
- Compact and lightweight
- Suitable for low-power IoT applications



Figure 5.6: Li-ion battery

4. Buck Converter Module

A DC-DC buck converter module shown in Figure 5.7, is used to regulate the battery voltage and provide a stable 5V supply for the STM32 microcontroller and LoRa communication module. The buck converter improves overall power efficiency by minimizing unnecessary power dissipation.

Functions:

- Voltage step-down regulation
- Stable 5V power supply generation
- Improved power efficiency
- Protection against voltage fluctuations

Advantages:

- High efficiency compared to linear regulators
- Reduced heat generation
- Stable operation for microcontroller systems



Figure 5.7: LM2596 Buck converter

5. AMS1117 Voltage Regulator

The AMS1117 voltage regulator module shown in Figure 5.8, provides a regulated 3.3V output supply for sensors and low-voltage components. Stable voltage regulation is important for ensuring accurate sensor readings and reliable operation.

Functions:

- Provides regulated 3.3V output
- Ensures stable sensor operation
- Reduces voltage noise and fluctuations
- Protects sensors from unstable supply conditions

Applications:

- DHT11 sensor supply
- Soil moisture sensor supply and rain sensor supply



Figure 5.8: AMS1117 Voltage regulator

6. LoRa Communication Module (SX1278)

The SX1278 LoRa module shown in Figure 5.9, provides long-range wireless communication between sensor nodes and the ESP32 gateway. LoRa technology is highly suitable for agricultural applications because it supports low-power communication over large distances with good reliability in open environments. The system implements time-staggered communication using RTC wake-up scheduling to prevent simultaneous packet transmission and reduce communication collisions between nodes.

Functions:

- Long-range wireless communication
- Low-power data transmission
- Reliable communication in agricultural fields
- Communication between sensor nodes and gateway

Advantages:

- Low power consumption
- Large communication range
- Better penetration in outdoor environments



Figure 5.9: SX1278 LoRa module

7. ESP32 Gateway

The ESP32 microcontroller shown in Figure 5.10, acts as the central gateway and processing unit of the entire system. It receives LoRa data from all sensor nodes, processes the received information, displays sensor readings locally, uploads data to the cloud server, and controls irrigation devices. The ESP32 was selected due to its built-in Wi-Fi capability, sufficient processing power, multiple GPIO interfaces, and compatibility with IoT applications.

Functions:

- Receives sensor data from LoRa nodes
- Processes incoming information
- Displays sensor data on LCD
- Uploads data to cloud server
- Controls relay module and irrigation devices
- Interfaces with mobile and cloud platforms



Figure 5.10: ESP32 Microcontroller

8. Wi-Fi Communication Unit

The ESP32 uses its built-in Wi-Fi module to establish internet connectivity and communicate with the backend server and cloud database. Wi-Fi communication enables remote monitoring and cloud-based storage of field information.

Functions:

- Internet connectivity
- Real-time cloud communication
- Data synchronization with server
- Remote monitoring support

Advantages:

- Wireless internet access
- Fast data transmission
- Easy integration with cloud platforms

9. Backend Server

The backend server developed using Node.js and Express acts as an intermediate communication layer between the ESP32 gateway and the cloud database. It receives sensor data through API requests, processes the information, and stores it in the database.

Functions:

- Receives HTTP POST requests
- Processes JSON sensor data
- Manages cloud communication
- Handles API endpoints
- Organizes and validates incoming data

Advantages:

- Scalable server architecture
- Fast processing capability
- Easy integration with IoT systems

10. Supabase Database

Supabase is used as the cloud database platform for storing sensor readings, node details, and system logs. It enables secure cloud storage and retrieval of agricultural data for real-time and historical monitoring.

Functions:

- Stores sensor information
- Maintains historical records
- Supports graphical trend analysis
- Enables dashboard data retrieval
- Stores system logs and alerts

Advantages:

- Cloud-based accessibility
- Real-time database synchronization
- Secure data storage
- Easy web integration

11. LCD Display (16×2 I2C)

The 16×2 I2C LCD display shown in Figure 5.11, provides local visualization of sensor data received from different nodes. The I2C communication interface reduces wiring complexity and simplifies microcontroller interfacing.

Functions:

- Displays node information
- Shows temperature, humidity, and moisture readings
- Provides local monitoring capability
- Displays sensor data sequentially

Advantages:

- Simple interface
- Low wiring requirement
- Easy readability in field conditions



Figure 5.11: LCD Display

12. Relay Module

The 4-channel relay module shown in Figure 5.12, acts as an electrical switching interface between the ESP32 and irrigation devices such as pumps and solenoid valves. Since the ESP32 operates at low voltage and current levels, relays are required to safely control higher-power devices.

Functions:

- Controls water pump operation
- Controls solenoid valve switching
- Provides electrical isolation
- Enables safe high-current switching

Advantages:

- Electrical isolation protection
- Reliable switching operation
- Easy interfacing with microcontrollers



Figure 5.12: 4-channel relay

13. Solenoid Valves

The solenoid valves shown in Figure 5.13, control water flow to different irrigation zones within the agricultural field. Each valve can independently regulate water supply to a specific section.

Functions:

- Controls water distribution
- Enables zone-wise irrigation
- Regulates irrigation flow
- Supports efficient water management

Advantages:

- Fast switching operation
- Automatic water flow control
- Reduced water wastage



Figure 5.13: 5V Solenoid valve

14. Water Pump

The 5V DC water pump shown in Figure 5.14, supplies water to the irrigation system whenever irrigation is activated. It works together with the relay module and solenoid valves to distribute water to the required field sections.

Functions:

- Supplies water for irrigation
- Maintains water flow in the system
- Supports irrigation operation

Advantages:

- Efficient water transfer
- Suitable for agricultural irrigation systems



Figure 5.14: 5V DC Motor pump

15. Blynk Mobile Application

The Blynk mobile application provides a user-friendly remote interface for farmers to monitor field conditions and manually control irrigation devices from a smartphone.

Functions:

- Remote irrigation control
- Mobile-based monitoring
- Real-time sensor visualization
- User-friendly interface

Advantages:

- Easy remote access
- Simple irrigation control
- Supports smart farming applications

CHAPTER - 6

IMPLEMENTATION OF “IoT Enabled Smart Sugarcane Farming System”

This chapter covers the implementation and integration of all hardware and software components of the proposed IoT Enabled Smart Sugarcane Farming System. It describes the setup and testing of STM32 sensor nodes, LoRa communication modules, ESP32 gateway, irrigation components, and the cloud-based monitoring platform. Real-world implementation details, communication setup, and software configuration are also presented.

6.1 Flowchart of IoT Enabled Smart Sugarcane Farming System

The flowchart shown in Figure 6.1 represents the overall working process of the proposed IoT Enabled Smart Sugarcane Farming System. The operation begins with the initialization of the STM32 sensor nodes and the ESP32 gateway. During initialization, all sensors, communication modules, and display units are configured for proper operation.

After initialization, the sensor nodes continuously collect environmental data such as temperature, humidity, soil moisture, and rainfall using the connected sensors. The STM32 microcontroller processes the collected sensor readings and prepares the data for wireless transmission.

The processed data is transmitted from the STM32 sensor nodes to the ESP32 gateway using SX1278 LoRa communication modules. LoRa technology enables reliable long-range communication with low power consumption, making it suitable for agricultural field deployment. The ESP32 gateway receives the sensor data from all nodes and displays the readings sequentially on the 16×2 I2C LCD display for local monitoring. The gateway then connects to the internet through Wi-Fi and uploads the received sensor data to the cloud server and Supabase database.

The web dashboard retrieves the stored information and displays live sensor readings, historical graphs, and field status information for remote monitoring. The system continuously checks whether any environmental parameter exceeds predefined threshold values. If abnormal conditions are detected, such as low soil moisture, high temperature, or rainfall detection, the

Design and Implementation of an IoT-Enabled Smart Sugarcane Farming System

system generates alerts and precautionary recommendations for the farmer. Based on the soil moisture condition and crop requirements, the farmer can manually control irrigation using the Blynk mobile application.



Figure 6.1: Flow chart of the design

When irrigation is required, the ESP32 activates the relay module to control the water pump and corresponding solenoid valve for water distribution. If irrigation is not required, the system continues monitoring field conditions continuously.

The complete process repeats periodically, enabling real-time agricultural monitoring, efficient water management, and remote accessibility for smart sugarcane farming.

6.2 Explanation of the Hardware Design and Implementation

The hardware implementation of the proposed IoT Enabled Smart Sugarcane Farming System consists of sensor nodes, communication modules, gateway unit, irrigation control components, and power management circuits. Each hardware component is integrated to support real-time monitoring, wireless communication, cloud connectivity, and irrigation management in agricultural environments. Figure 6.2 shows the block diagram of the hardware implementation of the system design.

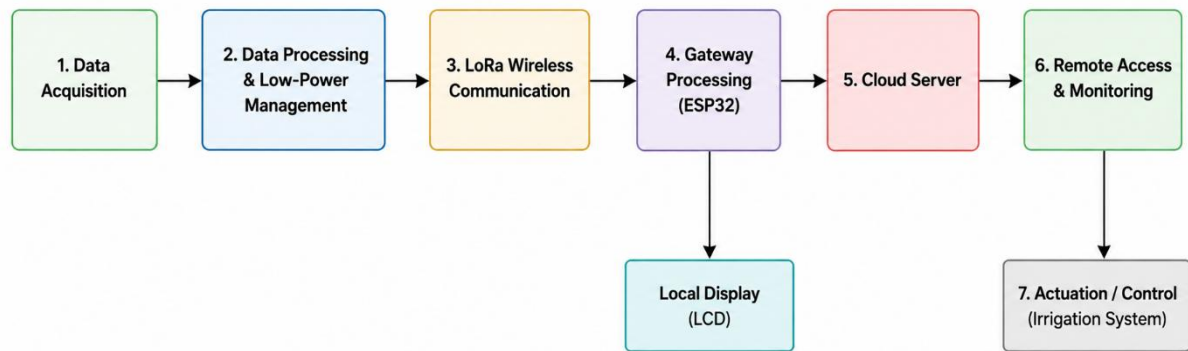


Figure 6.2: Block diagram of the hardware system design

6.2.1 Data Acquisition

The first stage of the proposed IoT Enabled Smart Sugarcane Farming System involves continuous acquisition of environmental and soil parameters from different locations in the

sugarcane field. Three individual STM32-based sensor nodes are deployed in the agricultural field for continuous monitoring of environmental and soil conditions.

Each sensor node interfaces with multiple sensors including the DHT11 sensor for temperature and humidity measurement, the capacitive soil moisture sensor for measuring soil water content, and the rain sensor for rainfall detection.

The DHT11 sensor is used for measuring environmental temperature and humidity conditions around the sugarcane field. Monitoring these parameters is important because temperature and humidity directly affect crop growth, soil evaporation, irrigation requirements, and overall agricultural productivity. The sensor contains an internal thermistor for temperature measurement and a capacitive humidity sensing element for detecting atmospheric moisture. It provides calibrated digital output values in degree Celsius ($^{\circ}\text{C}$) for temperature and relative humidity percentage (%RH) for humidity.

The DHT11 communicates with the STM32 microcontroller using a single-wire digital communication protocol, where both data transmission and reception occur through a single data pin. During operation, the STM32 sends a start signal to the DHT11 sensor, after which the sensor transmits digital temperature and humidity data to the controller.

The capacitive soil moisture sensor measures soil water content present in the agricultural field. The sensor provides analog voltage output proportional to the soil moisture level, which is read using the internal ADC (Analog-to-Digital Converter) of the STM32 microcontroller.

The rain sensor is used for detecting rainfall conditions in the agricultural field. The sensor operates on the principle of conductive rain detection, where water droplets falling on the sensor plate create conductive paths between exposed conductive tracks. When rainwater is present on the sensor surface, the electrical conductivity between the tracks increases, allowing the sensor to detect rainfall. The sensor module can provide both analog and digital outputs depending on the amount of water detected on the sensing plate. The STM32 continuously performs data acquisition by reading sensor values at predefined intervals.

6.2.2 Sensor Data Processing

The STM32 Nucleo F401RE microcontroller acts as the main processing and control unit in each sensor node of the proposed IoT Enabled Smart Sugarcane Farming System. The STM32 controller is selected due to its low power consumption, high processing performance, multiple communication interfaces, and suitability for embedded IoT applications.

After collecting the sensor readings, the STM32 controller processes the data and converts it into suitable engineering values such as:

- Temperature in degree Celsius (°C)
- Humidity percentage (%)
- Soil moisture percentage (%)
- Rain detection status (0 or 1)

The processed sensor information is then organized into structured data packets containing:

- Node ID
- Temperature
- Humidity
- Soil moisture
- Rain status

These packets are prepared for wireless transmission through the SX1278 LoRa module. In the proposed system, the STM32 periodically reads the sensor values, processes the data, and includes the readings in the LoRa transmission packet along with soil moisture and rain sensor information.

Soil Moisture Conversion Equation:

$$\text{Moisture (\%)} = \frac{(\text{Dry Value} - \text{ADC Value}) \times 100}{(\text{Dry Value} - \text{Wet Value})}$$

- ADC value decreases when moisture increases
- Calibration values are experimentally obtained
- Moisture is limited between 0–100%

6.2.3 Low-Power Management

To improve battery efficiency, the STM32 sensor nodes were implemented with RTC-based STOP mode functionality using STM32 HAL power management libraries. After collecting sensor readings and transmitting data through the SX1278 LoRa module, the STM32 controller enters STOP mode to minimize power consumption.

When the sensor nodes are powered ON, a time-staggered transmission mechanism is used to avoid communication collision between multiple LoRa nodes. The transmission sequence is as follows:

- **Node 1** → Transmits data immediately after startup and enters STOP mode
- **Node 2** → Waits for 6 seconds, transmits data, and enters STOP mode
- **Node 3** → Waits for 12 seconds, transmits data, and enters STOP mode

After entering STOP mode, all nodes use the RTC wake-up timer to wake up periodically every 60 seconds. Once awakened, each node reinitializes the required peripherals, acquires sensor data, transmits the data through the SX1278 LoRa module, and again enters STOP mode. Table 6.1 shows the RTC Wake-up Timing Model for each node.

This staggered startup delay ensures that the nodes do not transmit simultaneously during initialization, thereby reducing LoRa packet collision and improving communication reliability. The periodic STOP mode operation significantly reduces energy consumption and increases battery life for long-term agricultural field deployment.

UART debugging messages shown in Figure 6.3, were used to verify successful STOP mode entry and wake-up operation during testing.

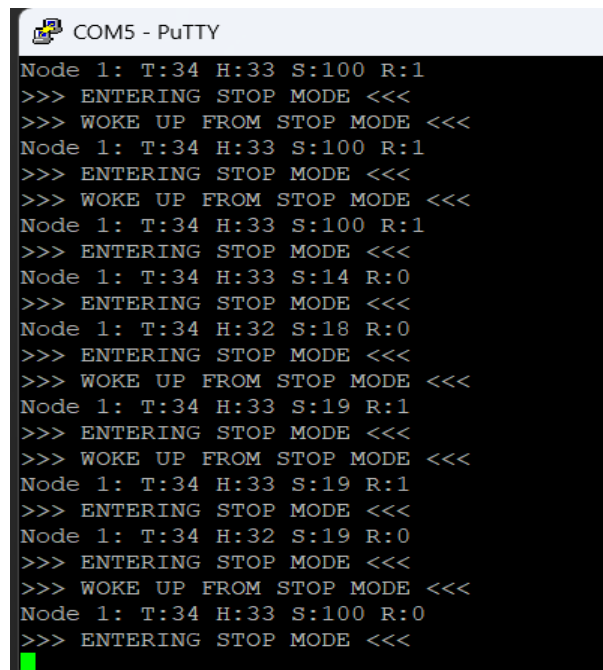
Example messages:

- “>>> ENTERING STOP MODE <<<”
- “>>> WOKE UP FROM STOP MODE <<<”

This implementation improves power efficiency, communication reliability, and overall system performance in remote smart agriculture environments.

Table 6.1 RTC Wake-up Timing model

Node	Initial Delay	Wake Interval
Node 1	0 s	60 s
Node 2	6 s	66 s
Node 3	12 s	72 s



```
COM5 - PuTTY
Node 1: T:34 H:33 S:100 R:1
>>> ENTERING STOP MODE <<<
>>> WOKE UP FROM STOP MODE <<<
Node 1: T:34 H:33 S:100 R:1
>>> ENTERING STOP MODE <<<
>>> WOKE UP FROM STOP MODE <<<
Node 1: T:34 H:33 S:100 R:1
>>> ENTERING STOP MODE <<<
Node 1: T:34 H:33 S:14 R:0
>>> ENTERING STOP MODE <<<
Node 1: T:34 H:32 S:18 R:0
>>> ENTERING STOP MODE <<<
>>> WOKE UP FROM STOP MODE <<<
Node 1: T:34 H:33 S:19 R:1
>>> ENTERING STOP MODE <<<
>>> WOKE UP FROM STOP MODE <<<
Node 1: T:34 H:33 S:19 R:1
>>> ENTERING STOP MODE <<<
Node 1: T:34 H:32 S:19 R:0
>>> ENTERING STOP MODE <<<
>>> WOKE UP FROM STOP MODE <<<
Node 1: T:34 H:33 S:100 R:0
>>> ENTERING STOP MODE <<<
```

Figure 6.3: RTC Based – Sleep modes

6.2.4 LoRa Wireless Communication Stage

The SX1278 LoRa module is used for long-range wireless communication between the STM32 sensor nodes and the ESP32 gateway. The module operates at a frequency of 433 MHz, which is suitable for low-power long-distance communication in open agricultural environments. The SX1278 communicates with the STM32 microcontroller using the SPI (Serial Peripheral Interface) communication protocol, where pins such as:

- MISO
- MOSI
- SCK
- NSS

are used for high-speed data exchange between the controller and the LoRa module. The STM32 processes sensor readings and forms structured data packets as shown in Table 6.2, containing:

- Node ID
- Temperature
- Humidity
- Soil moisture
- Rain status

The formatted packets are transmitted wirelessly through LoRa communication to the ESP32 gateway. LoRa technology provides several advantages such as:

- Long communication range
- Low power consumption
- Reliable data transmission

making it highly suitable for agricultural monitoring applications. In the proposed system, a collision avoidance mechanism is implemented using staggered RTC wake-up intervals to prevent simultaneous transmission from multiple sensor nodes.

- Node 1 transmits immediately after startup
- Node 2 and Node 3 transmit after predefined delays

This time-staggered communication reduces LoRa packet collision and improves communication reliability between the sensor nodes and the ESP32 gateway.

Table 6.2: Lora data packet

Field	Description
Node ID	Sensor node number
T	Temperature
H	Humidity
S	Soil Moisture
R	Rain Status

6.2.5 Gateway Processing

The ESP32 microcontroller acts as the central gateway and communication unit of the proposed IoT Enabled Smart Sugarcane Farming System. The ESP32:

- Receives LoRa data packets transmitted from all STM32 sensor nodes
- Processes incoming sensor information
- Parses the received data
- Displays readings locally
- Uploads sensor information to the cloud server

The received packets contain parameters such as:

- Node ID
- Temperature
- Humidity
- Soil moisture
- Rain status

After receiving the sensor data, the ESP32 parses the information and displays the readings locally on the 16×2 I2C LCD display for field monitoring purposes. The ESP32 also utilizes its built-in Wi-Fi capability to establish internet connectivity and communicate with the backend server. After processing the received sensor data, the ESP32:

- Creates JSON formatted packets
- Sends HTTP POST requests

Uploads sensor information to the server for cloud storage and remote monitoring

6.2.6 Local Display

A 16×2 LCD display with I2C interface is used in the proposed system for local visualization of sensor readings received from different sensor nodes. The I2C communication interface reduces the number of required microcontroller pins by using only two communication lines, namely:

- SDA (Serial Data)
- SCL (Serial Clock)

making the interfacing simpler and more efficient.

The LCD is connected to the ESP32 gateway, which continuously updates and displays sensor information. The LCD operates by sequentially displaying data received from all sensor nodes one after another.

Parameters such as:

- Node ID
- Temperature
- Humidity
- Soil moisture level
- Rain status

are shown on the display for quick field monitoring. This local display mechanism helps farmers and users observe real-time agricultural conditions directly at the gateway without requiring internet access or cloud connectivity.

6.2.7 Cloud Integration and Data Storage

The ESP32 gateway uploads sensor information to the backend server and Supabase cloud database using Wi-Fi communication.

- The cloud platform stores:
- Real-time sensor readings
- Historical records
- Node information
- System logs

REST APIs are used for communication between the ESP32 gateway and the cloud platform.

This cloud integration enables:

- Remote agricultural monitoring
- Historical data analysis
- Data visualization
- Cloud storage
- Real-time accessibility

6.2.8 Remote Monitoring and Control

A web-based dashboard is developed for real-time monitoring of agricultural conditions. The dashboard displays:

- Live sensor readings
- Historical data trends
- Node-wise monitoring
- Alerts and notifications
- Precautionary recommendations

The Blynk mobile application is integrated for irrigation control and remote accessibility.

Using the Blynk application, farmers can:

- Turn ON/OFF the water pump
- Control individual solenoid valves
- Perform remote irrigation management

based on soil moisture conditions and crop water requirements.

6.2.9 Irrigation Control

The irrigation control system consists of:

- 4-channel relay module
- 5V water pump
- Three 5V solenoid valves

The 4-channel relay module is used for controlling high-power irrigation devices such as the water pump and solenoid valves. Since the ESP32 operates at low voltage and current levels, the relay module acts as an interface between the low-power microcontroller circuit and the higher-power irrigation components. The relay module provides electrical isolation between the control circuit and the load circuit, protecting the ESP32 from voltage spikes and electrical damage during switching operations. The ESP32 controls the relay channels through its GPIO pins based on irrigation commands received from the Blynk mobile application.

When a relay channel is activated:

- The internal electromagnetic switch changes its contact position
- Current flows to the connected irrigation device
- The water pump and corresponding solenoid valve are activated

The proposed system uses three 5V solenoid valves and a 5V water pump for irrigation management. The water pump supplies water to the irrigation pipeline, while the solenoid valves control the flow of water to different sections of the sugarcane field. The implemented irrigation system supports zone-wise irrigation, where each solenoid valve controls water flow to a specific agricultural region monitored by an individual sensor node. Through the Blynk mobile application, the farmer can remotely activate the required valve and water pump according to soil moisture conditions and crop water requirements.

This selective irrigation mechanism:

- Improves water management efficiency
- Reduces unnecessary water wastage
- Prevents over-irrigation and waterlogging
- Supports efficient irrigation control in different field regions

6.2.10 Power Management Stage

The power supply circuit of the proposed system is designed to provide stable and efficient power for all sensor node components during field deployment. Each STM32 sensor node is powered using two 3.7V, 2400mAh Li-ion rechargeable batteries.

These batteries provide portable power supply for agricultural environments where direct electrical power may not be available. The batteries support long-duration operation and flexible deployment of sensor nodes in remote field locations. To ensure stable voltage levels for different hardware components, voltage regulation circuits are used in the system.

A DC-DC buck converter module is used to regulate the battery voltage and provide a stable 5V supply for:

- STM32 microcontroller
- SX1278 LoRa module

Additionally, an AMS1117 voltage regulator module is used to generate a stable 3.3V output for:

- DHT11 sensor
- Soil moisture sensor
- Rain sensor

These voltage regulation circuits:

- Reduce voltage fluctuations
- Improve power efficiency
- Ensure reliable operation of all system components
- Support continuous agricultural monitoring

The integration of RTC wake-up scheduling and low-power STOP mode further minimizes energy consumption and improves battery life for long-duration agricultural deployment.

6.3 Explanation of the Software Design and Implementation

The software design of the proposed IoT Enabled Smart Sugarcane Farming System is developed to support real-time monitoring, wireless communication, cloud integration, and irrigation control. The software architecture consists of embedded firmware for sensor nodes and gateway, backend server logic, cloud database management, and a web dashboard for remote monitoring. Figure 6.4 shows the block diagram of the hardware implementation of the system design.

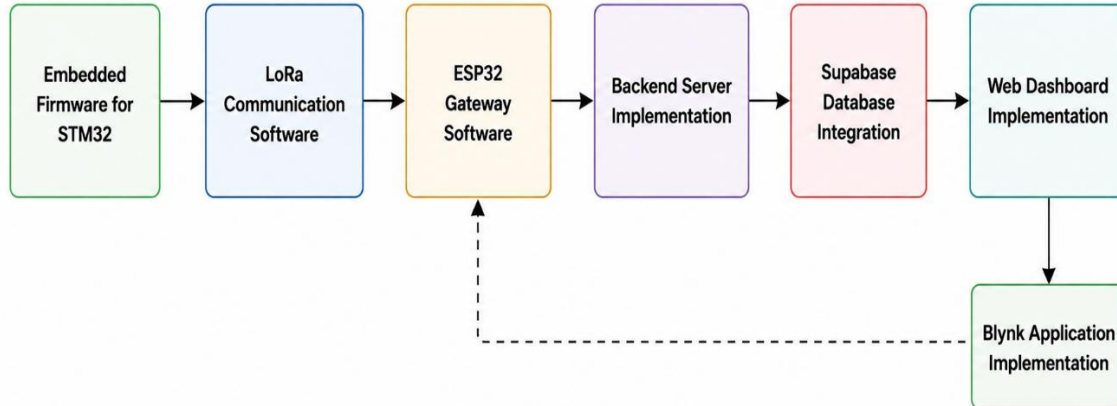


Figure 6.4: Block diagram of the software system design

6.3.1 Embedded Firmware for STM32

The embedded firmware for the STM32 Nucleo F401RE sensor nodes is developed using Embedded C to support sensor interfacing, data acquisition, LoRa communication, and low-power operation. During initialization, the firmware configures all connected peripherals including GPIO pins, ADC channels, SPI communication, UART debugging, and RTC modules. The capacitive soil moisture sensor output is read using the internal ADC of the STM32, while the DHT11 sensor values are acquired through single-wire digital communication. After collecting all sensor readings, the firmware processes the data and forms structured packets containing node ID, temperature, humidity, soil moisture, and rain status for LoRa transmission. To improve battery efficiency, the STM32 enters STOP mode after transmission and periodically wakes up using RTC interrupts for the next sensing cycle.

DHT11 Reading:

```
Rh_byte1 = DHT11_Read();  
Temp_byte1 = DHT11_Read();
```

ADC Reading:

```
HAL_ADC_Start(&hadc1);  
adc_sum += HAL_ADC_GetValue(&hadc1);
```

6.3.2 LoRa Communication Software

The LoRa communication software is implemented using the SX1278 LoRa module operating at 433 MHz frequency for long-range wireless communication between sensor nodes and the ESP32 gateway. The software configures SPI communication between the STM32 and LoRa module and handles wireless packet transmission and reception. Each transmitted packet contains a structured format including node ID, temperature, humidity, soil moisture level, and rain detection status. To prevent simultaneous transmission from multiple nodes, staggered transmission intervals are implemented where each node transmits after a predefined startup delay. This collision avoidance timing mechanism reduces LoRa packet collision and improves communication reliability within the agricultural monitoring network. Table 6.3 describes the communication parameters that needs to be set for both the transmitting and receiving LoRa, for reliable communication.

Table 6.3: Lora communication parameters

Parameter	Value
Frequency	433 MHz
Spreading Factor	SF7
Bandwidth	125 kHz
Coding Rate	4/5
Sync Word	0x34

6.3.3 ESP32 Gateway Software

The ESP32 gateway software is developed using Arduino IDE to perform central data processing, internet communication, and irrigation control functions. The ESP32 continuously receives LoRa packets transmitted from all STM32 sensor nodes and extracts the sensor readings from the received data packets. After parsing the information, the ESP32 creates JSON-formatted sensor data structures and establishes Wi-Fi connectivity for cloud communication. The gateway then sends HTTP POST requests containing sensor information to the backend server for cloud storage and remote monitoring. In addition, the ESP32 software also controls relay operations based on commands received from the Blynk mobile application.

Packet Parsing and Data Extraction code (ESP32)

```
sscanf(  
data.c_str(),  
"Node %d: T:%d H:%d S:%d R:%d", &nodeId, &temperature, &humidity, &soil, &rain);
```

6.3.4 Backend Server Implementation

The backend server is developed using Node.js and Express.js to manage communication between the ESP32 gateway and the cloud database. The server provides REST API endpoints that receive sensor data transmitted from the ESP32 through HTTP POST requests. Incoming JSON data is processed and validated to ensure correct parameter values before database storage. The backend server acts as an intermediate communication layer that organizes sensor information, handles data requests from the web dashboard, and supports reliable cloud integration for the smart farming system.

After processing received data, ESP32 sends sensor information to the backend server using **HTTP POST requests** in JSON format.

Example JSON structure:

```
{  
  "node_id": 1,  
  "temperature": 30,  
  "humidity": 72,  
  "soil_moisture": 45,  
  "rain": 0  
}
```

6.3.5 Supabase Database Integration

Supabase is used as the cloud database platform for storing real-time agricultural sensor data and system logs. The backend server securely connects to the Supabase database using API keys and database URLs configured through environment variables. The database stores parameters such as node ID, timestamp, temperature, humidity, soil moisture, and rain status

for each transmitted reading. Timestamp logging enables historical data storage and retrieval for trend analysis, graphical visualization, and long-term agricultural monitoring through the web dashboard.

Stored parameters include:

- Node ID
- Timestamp
- Temperature
- Humidity
- Soil moisture
- Rain status

This enables:

- Historical data storage
- Trend analysis
- Data retrieval for dashboard visualization

6.3.6 Web Dashboard Implementation

The web dashboard is developed to provide farmers with remote access to live agricultural field information through an internet-connected interface. The dashboard continuously retrieves sensor data from backend APIs and displays real-time updates for all sensor nodes. Graphical visualization tools are used to present live and historical trends of temperature, humidity, soil moisture, and rainfall conditions. The dashboard also includes alert generation features that notify users whenever environmental parameters exceed predefined threshold values, helping farmers make informed irrigation and field management decisions.

Dashboard features include:

- Real-time display of all 3 sensor nodes
- Historical graphs for environmental parameters
- Node-wise monitoring panels
- Alert notifications when thresholds are exceeded

6.3.7 Blynk Application Implementation

The Blynk mobile application is implemented to provide remote irrigation control and real-time monitoring through a smartphone interface. Virtual pins are configured within the Blynk platform to communicate with the ESP32 gateway over Wi-Fi. The application allows farmers to remotely activate or deactivate relay channels connected to the water pump and solenoid valves. Through this relay control mechanism, farmers can manually manage irrigation based on soil moisture conditions and crop water requirements without physically visiting the agricultural field.

CHAPTER - 7

RESULTS AND DISCUSSION

The proposed IoT Enabled Smart Sugarcane Farming System was successfully implemented and tested under real agricultural conditions at **VC Farms, University of Agricultural Sciences, Mandya**. The testing was carried out to evaluate the performance of sensor nodes, LoRa communication, gateway operation, cloud monitoring, and irrigation management. The experimental results confirmed reliable wireless communication, successful environmental monitoring, and efficient irrigation support for sugarcane cultivation.

7.1 Discussion of all the Individual and Overall Results

The complete system consisting of three STM32-based sensor nodes, SX1278 LoRa modules, ESP32 gateway, cloud monitoring dashboard, and irrigation control setup was successfully integrated and tested in the field environment at VC Farms, Mandya.



Figure 7.1 : Real-field deployment of three sensor nodes

Figure 7.1 shows the Real field implementation. Each sensor node successfully measured environmental parameters such as temperature, humidity, soil moisture, and rainfall using the DHT11 sensor, capacitive soil moisture sensor, and rain sensor. The STM32 microcontroller processed the sensor readings and transmitted the data wirelessly through the SX1278 LoRa communication module.

During testing, all three sensor nodes successfully transmitted sensor data to the ESP32 gateway without communication failure. The staggered transmission approach implemented

using RTC wake-up intervals helped prevent LoRa packet collision between multiple nodes and improved communication reliability.



(a)



(b)

Figure 7.2 : Testing of sensor nodes in different soil conditions: (a) Wet soil (b) Dry soil

Figure 7.2 shows the testing of sensor nodes in different soil conditions. LoRa communication system achieved a reliable communication range of approximately 50 meters in the agricultural field environment. Within this range, stable wireless communication was observed between the sensor nodes and the ESP32 gateway with minimal packet loss. The LoRa modules operated efficiently under outdoor farming conditions and enabled continuous data transmission from different locations in the field.

The ESP32 gateway successfully received sensor data from all nodes and displayed the readings sequentially on the I2C LCD display. The gateway also transmitted the received data to the cloud dashboard through Wi-Fi communication, enabling remote monitoring of field conditions.

The cloud dashboard successfully displayed live sensor values and historical data trends. Farmers could remotely observe field conditions and analyze environmental variations over time. Alert messages and recommendations were also generated whenever sensor values crossed predefined threshold conditions.

The irrigation system consisting of relay modules, solenoid valves, and water pump operated successfully through the Blynk mobile application. Irrigation was performed efficiently based

on soil moisture conditions observed from the sensor data. When soil moisture levels decreased, irrigation was manually activated through the mobile application to provide water to the required field section. This helped optimize water usage and avoid unnecessary irrigation.

The system demonstrated reliable performance under real agricultural conditions and validated the effectiveness of IoT-based monitoring and LoRa communication for smart sugarcane farming applications.

7.2 Explanation of the Results with Relevant Plots/Figures

The implemented system produced reliable and continuous monitoring results during field testing at VC Farms, Mandya. The sensor nodes continuously collected environmental and soil-related parameters and transmitted the data successfully to the ESP32 gateway using LoRa communication.

The successful reception of data from all three sensor nodes confirmed proper operation of the LoRa-based wireless sensor network. The ESP32 gateway displayed the received readings on the LCD display and uploaded the sensor data to the cloud dashboard for remote monitoring.

7.2.1 STM32 Sensor Node

The STM32 sensor node, shown in Figure 7.3 forms the primary sensing unit of the proposed IoT-enabled smart sugarcane farming system. Each node is developed using the STM32 Nucleo F401RE microcontroller integrated with a DHT11 sensor, capacitive soil moisture sensor, rain sensor, SX1278 LoRa communication module, and battery-powered supply circuitry. The node continuously monitors environmental and soil parameters from different regions of the agricultural field. The collected sensor values are processed by the STM32 controller and transmitted wirelessly through LoRa communication to the ESP32 gateway node. The sensor node is designed for low-power agricultural deployment by utilizing RTC wake-up scheduling and STOP mode operation, thereby improving battery life and enabling long-duration field monitoring.

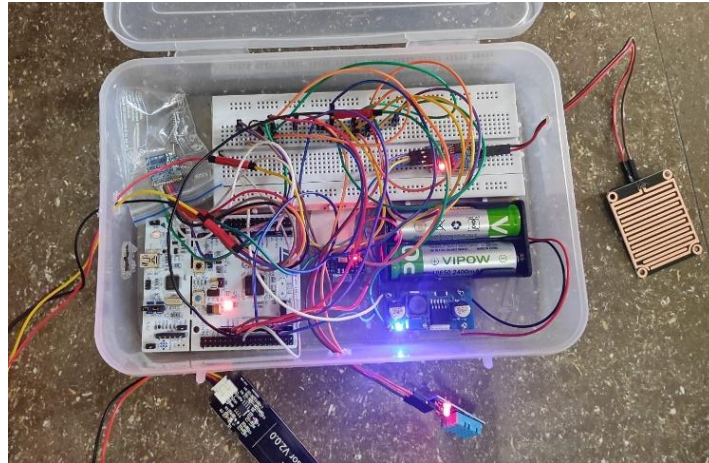


Figure 7.3 : STM32 Sensor node

7.2.2 ESP32 Gateway Node

The ESP32 gateway node, shown in Figure 7.4 acts as the central communication and processing unit of the proposed system. It receives sensor data transmitted from all STM32 sensor nodes through SX1278 LoRa communication modules operating at 433 MHz frequency. The ESP32 processes the received information, displays the data locally on the I2C LCD module, and forwards the sensor readings to the cloud server through Wi-Fi communication. The gateway also interfaces with the relay module, solenoid valves, and water pump for irrigation support. The implementation of the ESP32 gateway enables centralized monitoring, remote accessibility, and real-time cloud integration for agricultural field management.

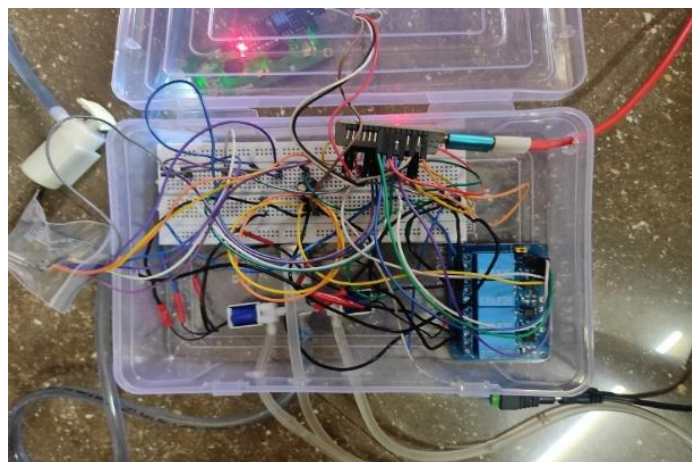


Figure 7.4 : ESP32 Gateway node

7.2.3 Serial Monitor Output Showing LoRa Communication and Data Transmission

The serial monitor output, shown in Figure 7.5 demonstrates successful LoRa-based wireless communication between the STM32 sensor nodes and the ESP32 gateway. The received packets contain environmental parameters such as temperature, humidity, soil moisture level, and rain sensor status from different nodes deployed in the sugarcane field. The output confirms reliable packet transmission, correct node identification, and successful data reception at the gateway unit. The results also validate the effectiveness of the time-staggered RTC wake-up scheduling technique implemented to avoid LoRa packet collisions between multiple sensor nodes operating simultaneously in the agricultural environment.

```
....
WiFi Connected
10.114.57.71
LoRa Ready
System Ready
Received: Node 1: T:52 H:16 S:14 R:0
{"node":"node1","temperature":52,"humidity":16,"soil_moisture":14,"rain":0}
HTTP Code: 201
{"success":true,"message":"Data saved successfully","data":{"id":126,"temperature":52,"humidity":16,"soil_moisture":14,"rain":0,"created_at":"2026-05-05T10:14:57.71Z"}}
Received: Node 2: T:50 H:21 S:15 R:0
{"node":"node2","temperature":50,"humidity":21,"soil_moisture":15,"rain":0}
HTTP Code: 201
{"success":true,"message":"Data saved successfully","data":{"id":127,"temperature":50,"humidity":21,"soil_moisture":15,"rain":0,"created_at":"2026-05-05T10:15:15.15Z"}}
Received: Node 3: T:39 H:39 S:10 R:0
{"node":"node3","temperature":39,"humidity":39,"soil_moisture":10,"rain":0}
HTTP Code: 201
{"success":true,"message":"Data saved successfully","data":{"id":128,"temperature":39,"humidity":39,"soil_moisture":10,"rain":0,"created_at":"2026-05-05T10:15:39.10Z"}}
Received: Node 1: T:53 H:16 S:14 R:0
{"node":"node1","temperature":53,"humidity":16,"soil_moisture":14,"rain":0}
HTTP Code: 201
{"success":true,"message":"Data saved successfully","data":{"id":129,"temperature":53,"humidity":16,"soil_moisture":14,"rain":0,"created_at":"2026-05-05T10:15:53.14Z"}}
Received: Node 2: T:50 H:21 S:15 R:0
{"node":"node2","temperature":50,"humidity":21,"soil_moisture":15,"rain":0}
HTTP Code: 201
{"success":true,"message":"Data saved successfully","data":{"id":130,"temperature":50,"humidity":21,"soil_moisture":15,"rain":0,"created_at":"2026-05-05T10:16:15.15Z"}}
```

Figure 7.5 : ESP32 Serial monitor

7.2.4 LCD Display – Gateway

The I2C LCD display connected to the ESP32 gateway, shown in Figure 7.6 provides local real-time visualization of the received sensor readings. The display sequentially shows information from different sensor nodes, including temperature, humidity, soil moisture percentage, and rainfall status. This local monitoring feature enables farmers or field operators

to directly observe field conditions without requiring internet connectivity or cloud access. The LCD display improves usability and provides immediate feedback regarding the operational status of the system.



Figure 7.6 : LCD Sensor node data display

7.2.5 Blynk Mobile Application for Irrigation Control

The Blynk mobile application, shown in Figure 7.7 provides a user-friendly interface for remote irrigation control and field monitoring. Through the application, farmers can manually operate the water pump and individual solenoid valves according to soil moisture conditions and crop water requirements. The mobile interface allows remote accessibility from any location connected to the internet, thereby reducing the need for continuous physical presence in the agricultural field. The implementation of the Blynk platform improves irrigation flexibility and supports efficient water management practices in sugarcane cultivation.

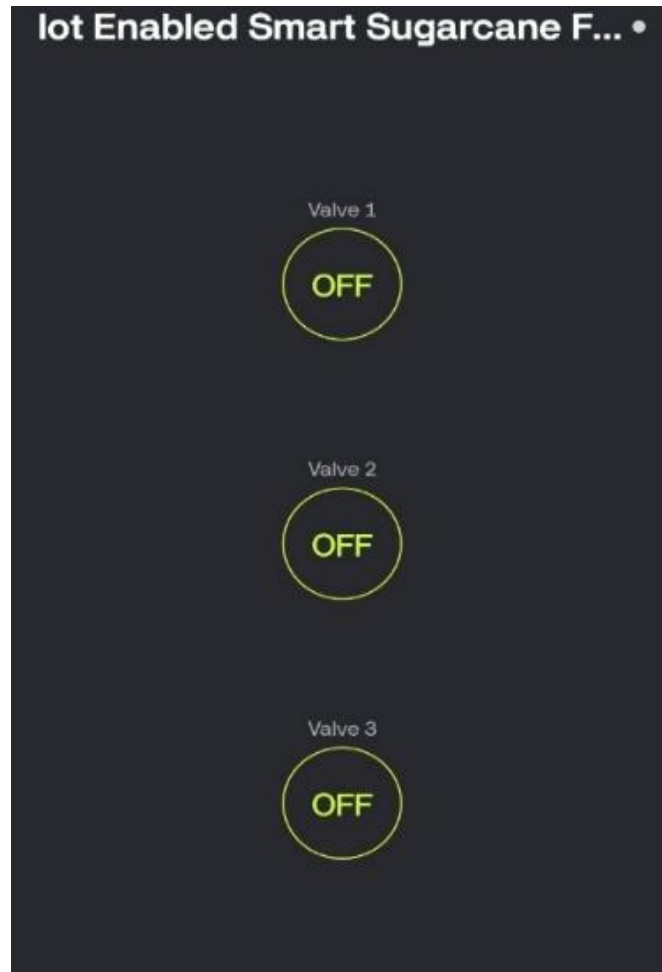


Figure 7.7 : Blynk mobile application

7.2.6 Web Dashboard

The developed web dashboard, shown in Figure 7.8 provides centralized monitoring and visualization of sensor data collected from all deployed nodes. The dashboard displays real-time environmental conditions including soil moisture, temperature, humidity, and rainfall information. The interface also supports historical data visualization and system status monitoring. By integrating cloud communication and remote accessibility, the dashboard enables farmers and researchers to analyze field conditions continuously and make informed irrigation decisions based on real-time sensor information.



Figure 7.8 : Web dashboard

7.2.7 Live Sensor Dashboard

The live readings and trend analysis dashboard provides real-time visualization of environmental parameters such as temperature, humidity, soil moisture, and rainfall collected from IoT sensor nodes. By continuously monitoring these readings, farmers can perceive changes and trends in field conditions over time, enabling informed decision-making for irrigation management, crop health monitoring, and precision farming practices.

Graph Parameter Representation:

Temperature (°C):

Represents the atmospheric temperature recorded by the sensor nodes. It helps in understanding the surrounding environmental conditions that affect crop growth and water evaporation rates.

Humidity (%):

Indicates the amount of moisture present in the air. Monitoring humidity helps in assessing weather conditions and predicting plant stress or disease possibilities.

Soil Moisture (%):

Measures the water content available in the soil. This parameter is essential for determining irrigation requirements and ensuring optimal water supply for sugarcane cultivation.

Rainfall (0 or 1):

Represents the detection of rainfall over a specific period. It helps farmers analyze natural water availability and reduce unnecessary irrigation.

Time Series / Trend Line:

Displays how sensor readings vary over time. It helps in identifying patterns, fluctuations, and long-term trends in field conditions for better agricultural planning.

Live Sensor Node Status:

Indicates the operational status and connectivity of each IoT sensor node deployed in the field, ensuring reliable real-time monitoring and data transmission.

Data Visualization Graphs:

The plotted graphs, shown in Figure 7.9 provide comparative analysis of multiple environmental parameters simultaneously, enabling users to easily perceive correlations between climate conditions and crop requirements.

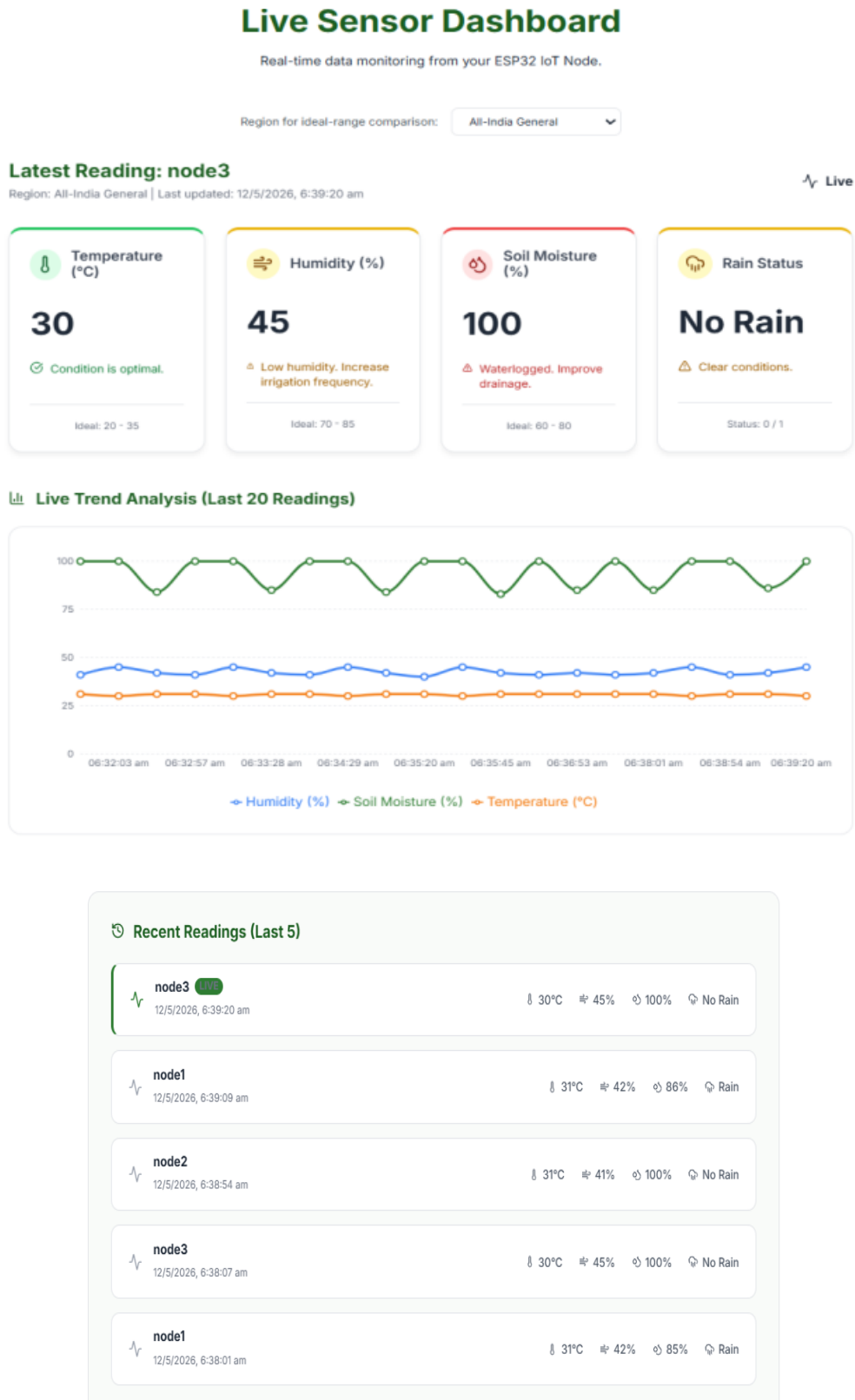


Figure 7.9 : Live Sensor dashboard

7.2.8 UI IoT Live Monitor

The IoT live monitoring user interface, shown in Figure 7.10 provides an organized visualization platform for remotely observing field conditions and irrigation status. The interface displays live sensor readings, node status, and communication updates in an easily understandable format. The successful operation of the monitoring interface confirms proper cloud integration, backend communication, and real-time synchronization between the deployed sensor nodes and remote monitoring platform. The UI improves accessibility and supports efficient decision-making for agricultural management.

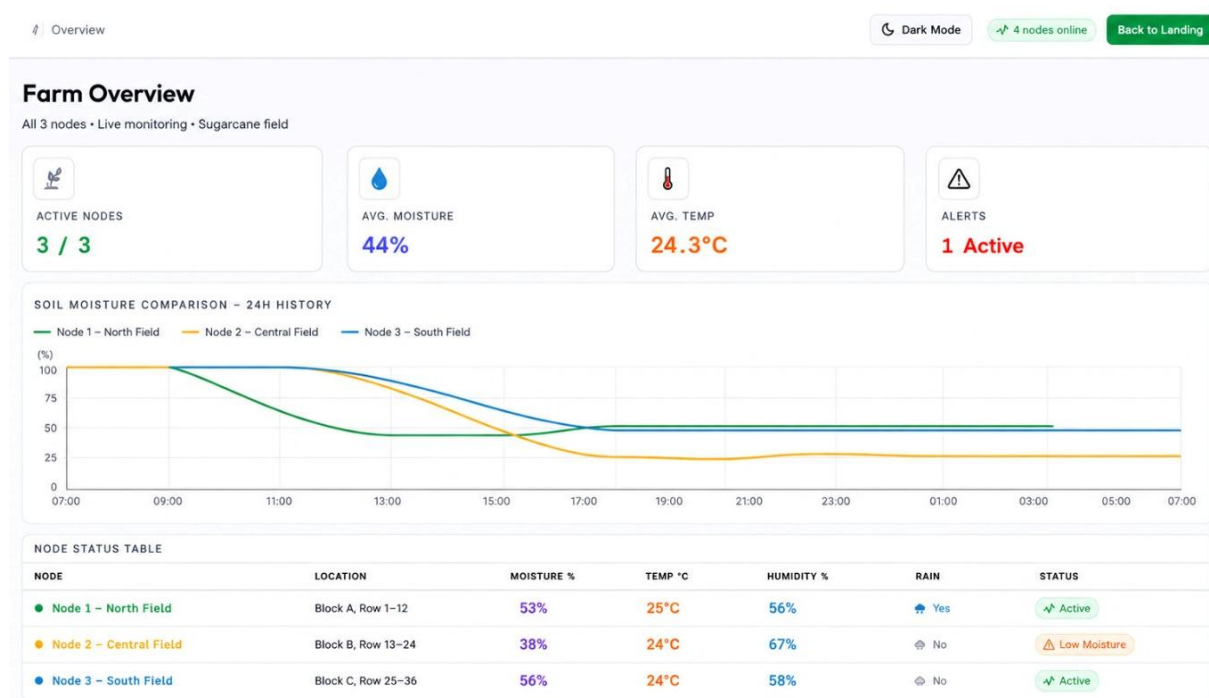
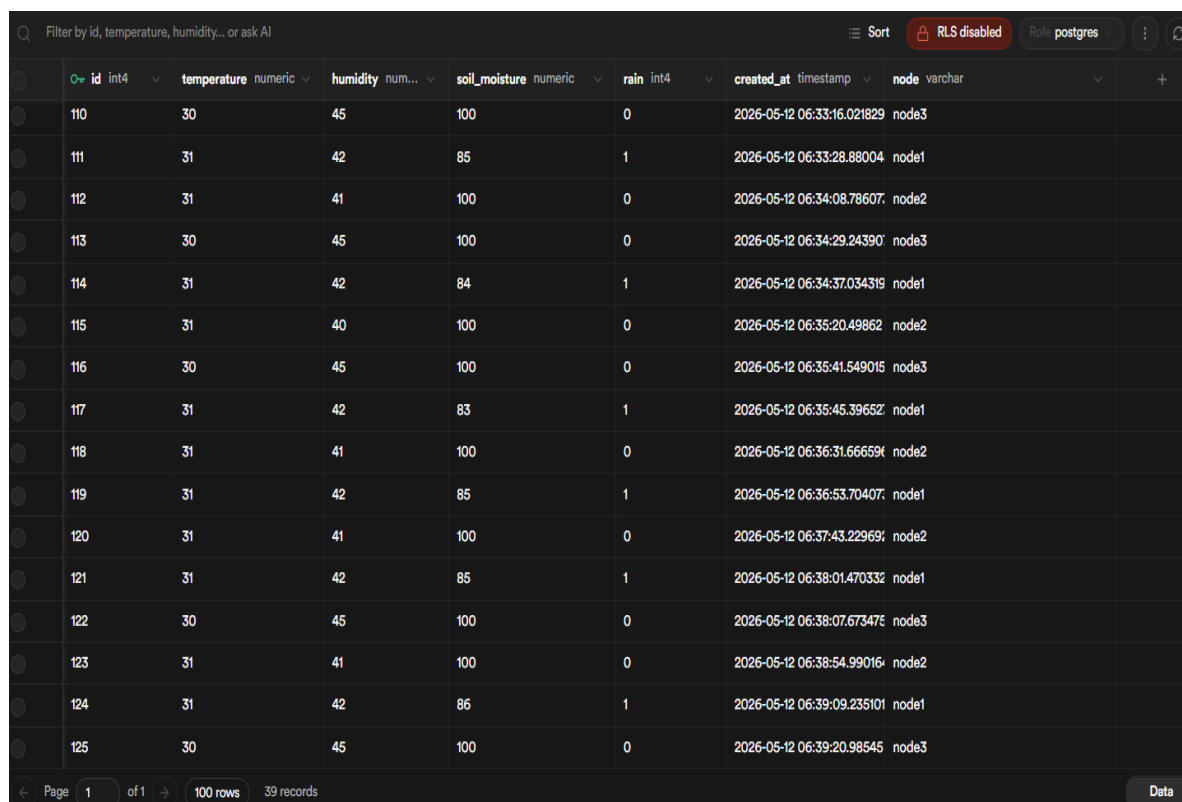


Figure 7.10 : UI IoT Live monitor

7.2.9 Supabase Cloud Database

The Supabase cloud database, shown in Figure 7.11 is used for secure storage and management of sensor data received from the ESP32 gateway. The database stores environmental readings, node information, timestamps, and irrigation-related logs for future analysis and monitoring. Cloud-based storage enables historical trend analysis, remote accessibility, and reliable data management for the proposed smart farming system. The successful integration of the

Supabase database validates the capability of the system to support scalable IoT-based agricultural monitoring applications.



The screenshot displays the Supabase database interface with a table containing 39 records. The table has the following columns: id (int4), temperature (numeric), humidity (numeric), soil_moisture (numeric), rain (int4), created_at (timestamp), and node (varchar). The data shows readings from three different nodes (node1, node2, node3) over time, with values for temperature, humidity, soil moisture, and rain.

id	temperature	humidity	soil_moisture	rain	created_at	node
110	30	45	100	0	2026-05-12 06:33:16.021829	node3
111	31	42	85	1	2026-05-12 06:33:28.88004	node1
112	31	41	100	0	2026-05-12 06:34:08.78607	node2
113	30	45	100	0	2026-05-12 06:34:29.24390	node3
114	31	42	84	1	2026-05-12 06:34:37.034319	node1
115	31	40	100	0	2026-05-12 06:35:20.49862	node2
116	30	45	100	0	2026-05-12 06:35:41.549015	node3
117	31	42	83	1	2026-05-12 06:35:45.39652	node1
118	31	41	100	0	2026-05-12 06:36:31.66659	node2
119	31	42	85	1	2026-05-12 06:36:53.70407	node1
120	31	41	100	0	2026-05-12 06:37:43.22969	node2
121	31	42	85	1	2026-05-12 06:38:01.470332	node1
122	30	45	100	0	2026-05-12 06:38:07.673475	node3
123	31	41	100	0	2026-05-12 06:38:54.99016	node2
124	31	42	86	1	2026-05-12 06:39:09.235101	node1
125	30	45	100	0	2026-05-12 06:39:20.98545	node3

Figure 7.11 : Supabase cloud database

7.3 Environmental Sensor Data Analysis

Environmental monitoring was performed using three STM32-based sensor nodes deployed at different locations in the sugarcane field. Each node continuously measured temperature, humidity, and soil moisture parameters from 09:00 AM to 01:30 PM. The collected data was transmitted through LoRa communication to the ESP32 gateway and uploaded to the cloud dashboard for monitoring and analysis. The recorded sensor readings from all three nodes demonstrate the capability of the proposed IoT-enabled smart farming system to monitor environmental variations across different regions of the agricultural field in real time.

7.3.1 Analysis of Environmental Trends

Node 1 Environmental Analysis:

Node 1 recorded temperature values ranging from 27°C to 33°C during the monitoring period from 09:00 AM to 01:30 PM as shown in Table 7.1 below. A gradual increase in temperature was observed throughout the day due to increased solar radiation and environmental heat during afternoon hours. Humidity values varied between 53% and 59%, indicating moderate atmospheric moisture conditions during field testing.

The soil moisture readings for Node 1 ranged from 46% to 52%. A gradual reduction in soil moisture was observed during higher temperature conditions due to evaporation and soil water absorption. However, the moisture levels remained within acceptable limits for sugarcane cultivation, indicating adequate irrigation conditions in that section of the field.

Overall, Node 1 demonstrated stable environmental conditions with balanced moisture availability and moderate temperature variation suitable for healthy crop growth.

Table 7.1: Real-time readings of soil and environmental parameter sensors for node1

Time	Temperature (°C)	Humidity (%)	Soil Moisture (%)
09:00 AM	27	56	52
09:30 AM	27	54	51
10:00 AM	28	58	51
10:30 AM	28	53	50
11:00 AM	30	56	48
11:30 AM	30	55	46
12:00 PM	31	55	46
12:30 PM	31	55	47
01:00 PM	32	59	46
01:30 PM	33	54	47

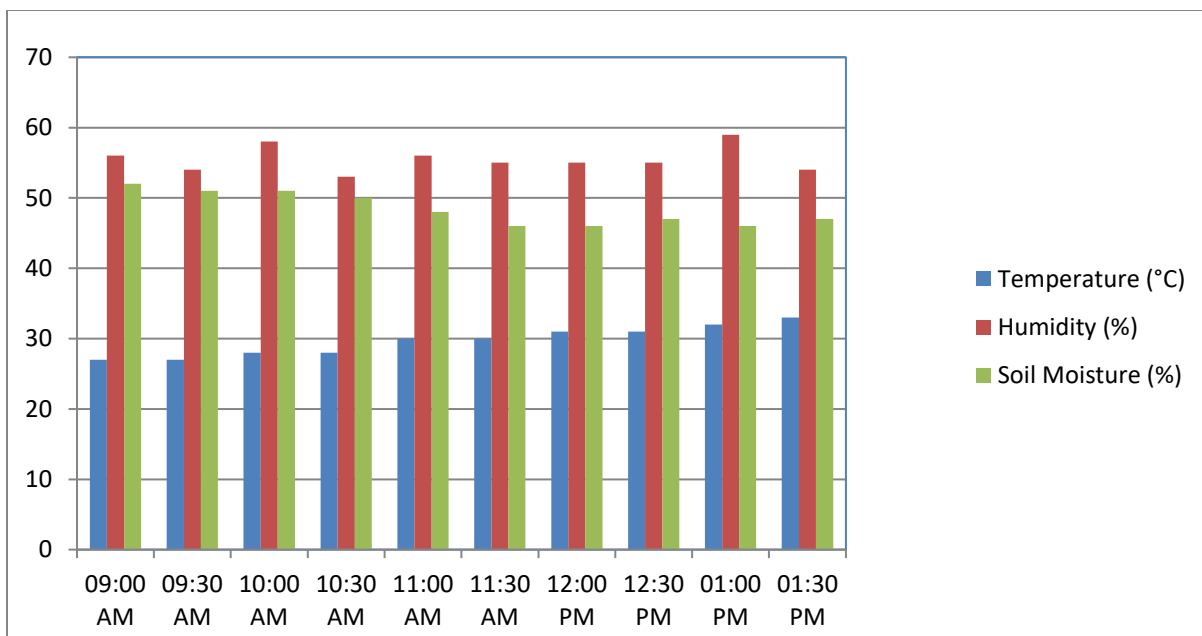


Figure 7.12 : Bar graph of the sensor readings for node 1

Node 2 Environmental Analysis:

Node 2 recorded temperature values ranging from 27°C to 33°C during the monitoring period, as shown in Table 7.2 below. Similar to Node 1, the temperature gradually increased during afternoon hours because of increased sunlight exposure and atmospheric heating. Humidity values fluctuated between 55% and 61%, representing comparatively higher atmospheric moisture conditions among the three nodes.

The soil moisture values ranged between 25% and 34%, which were significantly lower compared to Node 1 and Node 3. The moisture levels gradually decreased during daytime hours, indicating comparatively dry soil conditions and higher moisture loss in that region of the field. These readings suggest that additional irrigation support may be required in this section to maintain proper soil moisture for sugarcane cultivation.

The observed variations highlight the importance of multi-node environmental monitoring for accurate region-wise irrigation management.

Table 7.2: Real-time readings of soil and environmental parameter sensors for node 2

Time	Temperature (°C)	Humidity (%)	Soil Moisture (%)
09:00 AM	27	58	33
09:30 AM	27	60	34
10:00 AM	28	55	32
10:30 AM	28	61	33
11:00 AM	30	60	30
11:30 AM	30	55	26
12:00 PM	31	56	26
12:30 PM	31	59	25
01:00 PM	32	56	26
01:30 PM	33	61	26

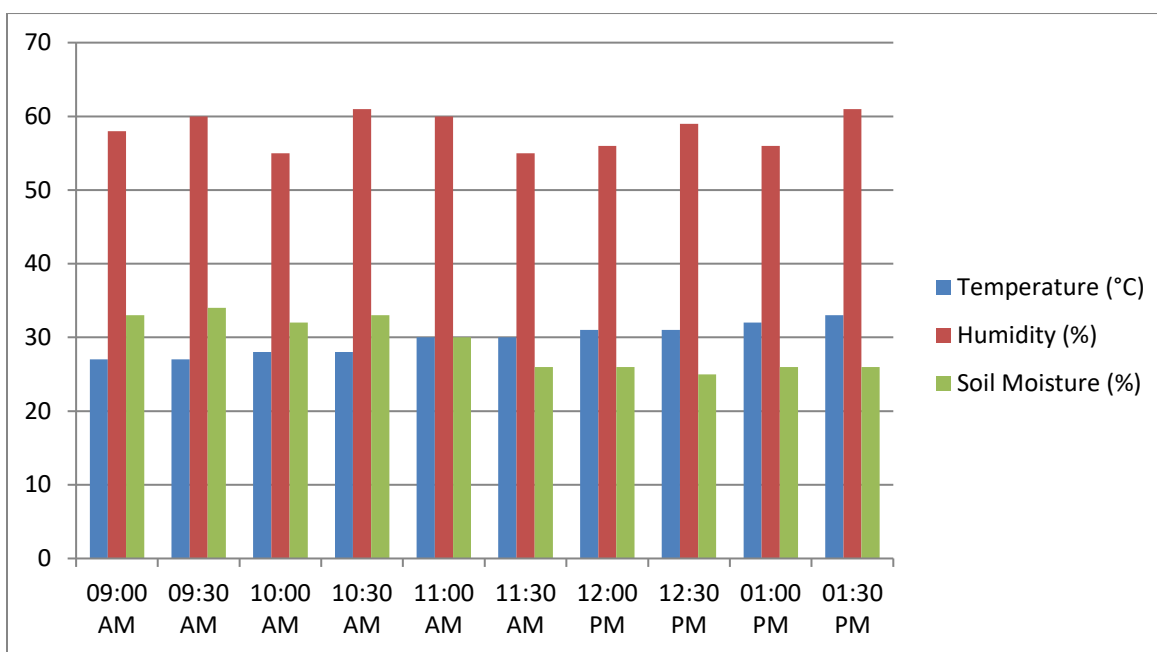


Figure 7.13 : Bar graph of the sensor readings for node 2

Node 3 Environmental Analysis:

Node 3 recorded temperature values between 27°C and 33°C throughout the testing period, as shown in Table 7.3 below. The gradual increase in temperature during afternoon hours closely matched the environmental trend observed in the other nodes. Humidity values varied from 51% to 57%, while soil moisture values ranged from 45% to 50%.

Design and Implementation of an IoT-Enabled Smart Sugarcane Farming System

The soil moisture readings in Node 3 remained comparatively stable during the monitoring period, although slight reductions were observed during increased temperature conditions due to evaporation effects. The environmental conditions recorded by Node 3 indicate sufficient moisture availability and moderate temperature levels suitable for sugarcane farming. Compared to Node 2, Node 3 maintained better soil moisture retention, indicating comparatively improved irrigation conditions in that region of the field.

Table 7.3: Real-Time readings of soil and environmental parameter sensors for node 3

Time	Temperature (°C)	Humidity (%)	Soil Moisture (%)
09:00 AM	27	56	50
09:30 AM	27	52	49
10:00 AM	28	57	48
10:30 AM	28	56	49
11:00 AM	30	56	50
11:30 AM	30	51	48
12:00 PM	31	51	47
12:30 PM	31	55	46
01:00 PM	32	52	45
01:30 PM	33	57	45

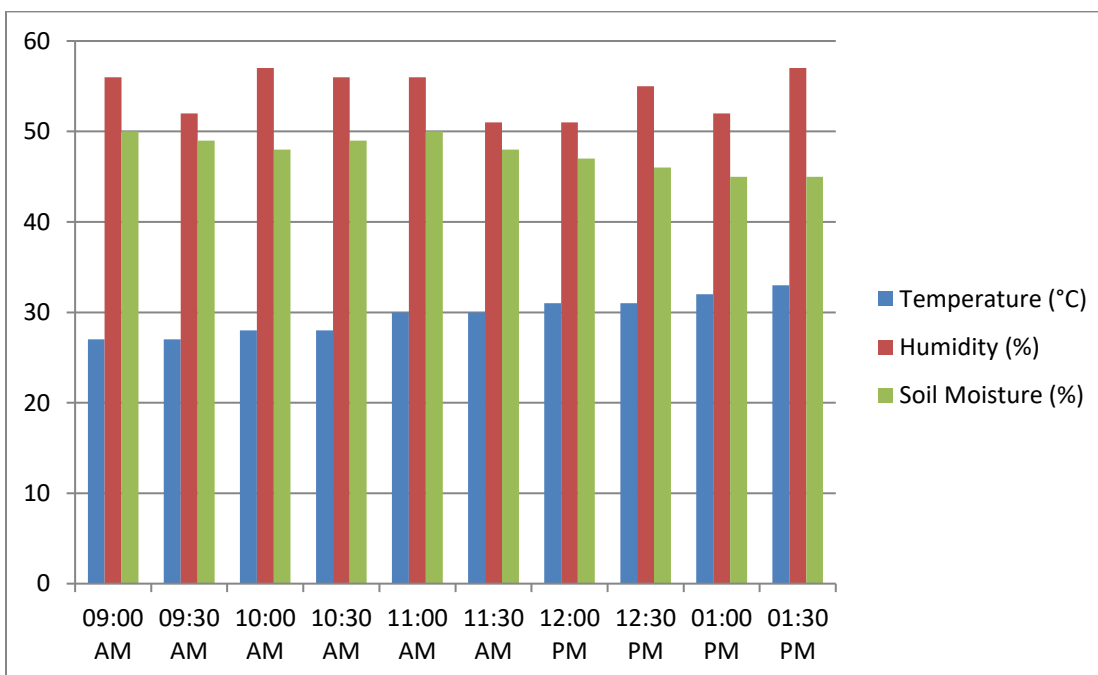


Figure 7.14 : Bar graph of the sensor readings for node 3

7.3.2 Comparative Analysis of All Sensor Nodes:

The environmental analysis of all three sensor nodes demonstrates noticeable variations in temperature, humidity, and soil moisture conditions across different regions of the sugarcane field. All three nodes recorded gradual increases in temperature during daytime hours, with the maximum temperature reaching 33°C at 01:30 PM due to increased solar radiation and atmospheric heating.

Humidity exhibited an inverse relationship with temperature variation. As temperature increased, humidity showed slight reductions because of increased evaporation and reduced atmospheric moisture retention. The humidity values remained within moderate ranges suitable for sugarcane cultivation throughout the testing duration.

Significant variations were observed in soil moisture levels among the deployed sensor nodes. Node 1 and Node 3 maintained comparatively higher soil moisture levels between 45% and 52%, indicating better moisture retention and irrigation conditions. In contrast, Node 2 consistently recorded lower soil moisture values between 25% and 34%, indicating comparatively dry soil conditions requiring additional irrigation attention.

Analysis of Environmental Trends:

The bar graphs in Figures 7.12, 7.13, and 7.14 illustrate the variation of temperature, humidity, and soil moisture levels recorded from the three sensor nodes during the monitoring period from 09:00 AM to 01:30 PM.

The temperature gradually increased during morning and afternoon hours across all three sensor nodes due to increasing solar radiation intensity and environmental heat exposure. The maximum temperature of 33°C was observed at 01:30 PM in all nodes.

Humidity followed an inverse trend compared to temperature variation. As atmospheric temperature increased, the relative humidity showed slight reductions because of increased evaporation and atmospheric heating. During lower temperature intervals, humidity levels slightly improved due to reduced evaporation effects.

Soil moisture levels gradually decreased during daytime hours because of continuous evaporation and water absorption by the soil. Node 2 recorded the lowest soil moisture values, indicating comparatively dry soil conditions, while Node 1 and Node 3 maintained relatively higher moisture levels throughout the monitoring period.

The observed environmental trends closely match practical agricultural conditions in sugarcane cultivation, thereby validating the effectiveness, reliability, and accuracy of the developed IoT Enabled Smart Sugarcane Farming System for real-time environmental monitoring and irrigation support applications.

Comparison with Expected Agricultural Conditions:

Table 7.4 : Comparison of observed environmental trends with expected agricultural behaviour

Parameter	Observed Behaviour	Expected Agricultural Behaviour	Result
Temperature	Increased during afternoon hours	Expected due to solar heating	Matched
Humidity	Decreased with temperature rise	Expected inverse relationship	Matched
Soil Moisture	Reduced during daytime	Expected due to evaporation	Matched
Evening Recovery	Moisture and humidity increased	Expected due to cooling conditions	Matched

The obtained results closely match the expected environmental trends observed in agricultural fields, as shown in the above Table 7.4. The inverse relationship between temperature and humidity and the gradual reduction in soil moisture during daytime validate the accuracy of the deployed sensors and monitoring architecture.

The experimental observations confirm that the developed IoT-enabled Smart Sugarcane Farming System successfully captures real-time environmental variations under practical field

conditions. The reliable monitoring performance demonstrates the suitability of the proposed system for smart irrigation planning and precision agriculture applications.

System Performance Validation:

The developed IoT-enabled Smart Sugarcane Farming System was successfully tested under real agricultural field conditions to evaluate its monitoring, communication, and cloud integration performance. The system continuously acquired environmental data from multiple sensor nodes and reliably transmitted the information to the ESP32 gateway using LoRa wireless communication.

The observed LoRa communication range was approximately 50 meters under field conditions. Stable communication was maintained within this range, demonstrating the suitability of SX1278 LoRa modules for short-to-medium range agricultural monitoring applications.

The ESP32 gateway successfully processed the received sensor data and uploaded it to the cloud database through Wi-Fi connectivity. Real-time sensor readings were displayed on the web dashboard and mobile application without significant communication delay. The implemented system also generated live environmental monitoring data useful for irrigation planning and field condition analysis.

The experimental results confirmed the successful operation of the following functionalities:

- Continuous environmental monitoring
- Real-time sensor data acquisition
- Reliable LoRa-based wireless communication
- Cloud database storage using Supabase
- Mobile-based irrigation monitoring and control
- Low-power sensor node operation using RTC-based sleep modes

The obtained results validate that the proposed system provides an efficient and reliable solution for smart sugarcane farming applications. The developed architecture supports precision agriculture by improving irrigation management, reducing manual monitoring effort, and enabling efficient utilization of water resources.

CHAPTER - 8

CONCLUSION AND FUTURE SCOPE

This chapter presents the overall conclusion of the proposed IoT Enabled Smart Sugarcane Farming System and discusses possible future improvements. The developed system successfully demonstrated real-time agricultural monitoring, long-range wireless communication, cloud-based data management, and irrigation support for smart farming applications. Possible future enhancements for improving automation, communication range, intelligence, and scalability are also discussed.

8.1 Conclusion of the Project

The IoT Enabled Smart Sugarcane Farming System was successfully designed, implemented, and tested for real-time agricultural monitoring and irrigation support. The developed system utilized STM32-based sensor nodes integrated with temperature, humidity, soil moisture, and rain sensors to continuously monitor field conditions at different locations in the sugarcane field.

LoRa communication using SX1278 modules enabled reliable wireless transmission of sensor data from all sensor nodes to the ESP32 gateway. During field testing at VC Farms, Mandya, all three sensor nodes successfully transmitted data to the gateway with a communication range of approximately 50 meters. The staggered LoRa transmission scheduling approach effectively prevented packet collision and improved communication reliability.

The ESP32 gateway successfully displayed sensor readings on the I2C LCD display and transmitted the data to the cloud dashboard using Wi-Fi communication. The web dashboard enabled remote monitoring of field conditions, historical data analysis, and generation of smart alerts and recommendations.

The irrigation control system using relay modules, solenoid valves, and water pump operated successfully through the Blynk mobile application. Irrigation decisions were made efficiently

based on real-time soil moisture conditions, helping improve water management and reduce unnecessary water usage.

The implementation of RTC-based STOP mode significantly reduced power consumption in STM32 sensor nodes, making the system suitable for long-term battery-powered agricultural deployment.

Overall, the proposed system demonstrated that IoT technology, wireless sensor networks, cloud integration, and mobile-based irrigation control can effectively support smart agriculture and improve monitoring efficiency in sugarcane farming applications.

8.2 Scope for Future Work

Although the proposed system achieved successful real-time monitoring and irrigation support, several improvements and advanced features can be incorporated in future developments.

Future work may include implementing fully automatic irrigation control based on real-time soil moisture thresholds without requiring manual intervention through the Blynk application. Artificial Intelligence (AI) and Machine Learning (ML) algorithms can also be integrated for predictive irrigation scheduling, crop health analysis, and environmental prediction.

The communication range can be improved further by using advanced LoRaWAN infrastructure or high-gain antennas for large-scale agricultural deployment. Additional sensors such as pH sensors, nutrient sensors, light intensity sensors, and weather monitoring modules can be integrated to provide more comprehensive agricultural analysis.

Solar-powered energy systems can be implemented to improve energy sustainability and eliminate dependency on battery replacement. Edge computing techniques may also be incorporated to process sensor data locally and reduce cloud dependency and communication latency.

Future versions of the system can include GPS-based node tracking, mobile notifications, multilingual user interfaces, and advanced analytics dashboards for improved farmer

accessibility and usability. Integration with government agricultural platforms and smart farming ecosystems can further enhance large-scale agricultural management and automation. The proposed system provides a strong foundation for future smart agriculture solutions and demonstrates the potential of IoT technologies in improving agricultural productivity, water management, and sustainable farming practices.

CHAPTER - 9

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Mapping of Undergraduate Final year Projects to Sustainable Development Goals (SDG)

Department	Electronics and Telecommunication
Project Title	Design and Implementation of an IoT-Enabled Smart Sugarcane Farming System
Name of the Project Guide	Dr. S G Shivaprasad Yadav

Goal No.	Sustainable Development Goal (SDG)	Mapping
1	No poverty	
2	Zero hunger	P
3	Good health and well-being	
4	Quality education	
5	Gender equality	
6	Clean water and sanitation	S
7	Affordable and clean energy	
8	Decent work and economic growth	
9	Industry, innovation and infrastructure	
10	Sustainable cities and communities	
11	Responsible consumption and production	
12	Reduced inequalities	
13	Climate action	S
14	Life below water	
15	Life on land	
16	Peace, justice, and strong institutions	
17	Partnerships for the goals	

Justification for SDG Mapping

Primary SDG – Goal 2: Zero Hunger (P)

The proposed IoT-Enabled Smart Sugarcane Farming System directly contributes to sustainable agriculture and improved crop productivity by continuously monitoring important environmental parameters such as soil moisture, temperature, humidity, and rainfall. The system helps farmers make informed irrigation decisions, reduce crop stress, and improve sugarcane yield through real-time monitoring and smart irrigation support. Efficient farming practices supported by IoT technology can enhance agricultural productivity, reduce resource wastage, and support food security, thereby aligning strongly with SDG 2 – Zero Hunger.

Secondary SDG – Goal 6: Clean Water and Sanitation (S)

The project promotes efficient water management by monitoring soil moisture conditions and enabling controlled irrigation based on real-time field data. Traditional irrigation methods often result in excessive water wastage and waterlogging. The proposed system helps optimize water utilization by allowing farmers to irrigate only when required, thereby conserving water resources and improving irrigation efficiency. The implementation of smart irrigation support contributes toward sustainable water usage and aligns with SDG 6 – Clean Water and Sanitation.

Secondary SDG – Goal 13: Climate Action (S)

The proposed system supports sustainable and climate-resilient agricultural practices by utilizing low-power IoT technologies, real-time environmental monitoring, and efficient resource management. Continuous monitoring of environmental conditions helps farmers respond effectively to changing weather patterns and climate-related agricultural challenges. The use of low-power LoRa communication and optimized irrigation techniques reduces unnecessary energy and water consumption, contributing to environmentally sustainable farming practices. Therefore, the project aligns with SDG 13 – Climate Action.